

RADIOMIC ANALYSIS OF CT SCANS IN COPD PATIENTS

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true

1. Radiomic features

Radiomic features for all patients were saved as a SummarizedExperiment

```
# import and visualize data:
load("./Data/radar_L1_Norm_scaled_adjusted_min_int_center_final.rda")
rownames(rdr_L1_final) <- sub(".original", "", rownames(rdr_L1_final))

rdr_L1_final
```

```
class: SummarizedExperiment
dim: 101 1773
metadata(1): extractor
assays(2): values adjusted_min_int_center
rownames(101): Elongation Flatness ... Complexity Strength
rowData names(4): feature_name image_type feature_description
                 feature_type
colnames(1773): 231400000005 231400000006 ... 26658003992 26658003997
colData names(201): filename sample_id ... change.fv1 D_SUBJID
```

Radiomic feature analysis

```
# Transpose radiomic data
features_adj <- t(assay(rdr_L1_final, "adjusted_min_int_center"))

library(ggplot2)
library(tidyr)
library(dplyr)
library(patchwork)

# Escalar cada feature (z-score)
features_scaled <- as.data.frame(scale(features_adj))

# Pasar a formato Largo
df_long <- features_scaled %>%
  mutate(id = row_number()) %>%
  pivot_longer(-id, names_to = "feature", values_to = "value")

# Número de features por panel
n_per_plot <- 20
features <- unique(df_long$feature)
feature_groups <- split(features, ceiling(seq_along(features) / n_per_plot))

# Crear Los plots
plots <- lapply(feature_groups, function(feats) {
  ggplot(df_long %>% filter(feature %in% feats),
    aes(x = feature, y = value)) +
    geom_violin(
      fill = "skyblue", color = "black",
      scale = "width", trim = FALSE,
      bw = 0.6      # <-- menor valor = más detalle; mayor = más suave (ajusta entre 0.3-0.
6)
    ) +
    coord_cartesian(ylim = c(-5, 5)) + # eje y entre -5 y 5
    theme_bw() +
    theme(
      axis.text.x = element_text(angle = 45, hjust = 1, size = 8),
      panel.grid.minor = element_blank()
    ) +
    labs(x = NULL, y = "Scaled value (z-score)")
})

png("radioviolin1.png",width = 3000, height = 4000,res = 400)
plots[[1]]
dev.off()
```

png
2

```
png("radioviolin2.png",width = 3000, height = 4000,res = 400)
plots[[2]]
dev.off()
```

```
png
2
```

```
png("radioviolin3.png",width = 3000, height = 4000,res = 400)
plots[[3]]
dev.off()
```

```
png
2
```

```
png("radioviolin4.png",width = 3000, height = 4000,res = 400)
plots[[4]]
dev.off()
```

```
png
2
```

```
png("radioviolin5.png",width = 3000, height = 4000,res = 400)
plots[[5]]
dev.off()
```

```
png
2
```

#Pearson distance:

```
results_p = ConsensusClusterPlus(
  features_adj,
  maxK = 20, # max cluster counts evaluated
  reps = 1000, # resamplings
  pItem = 0.8, # 80% item (feature) resampling
  pFeature = 0.8, # 80% feature resampling
  title = "./Data/CCP_Pearson",
  clusterAlg = "hc", # agglomerative hierarchical clustering algorithm
  distance = "pearson", #1-pearson correlation distances
  seed= 290597, finalLinkage = "ward.D2",
  innerLinkage = "ward.D2",
  plot = "png") # graphical results
```

Radiomic feature clusters

```
# Consensus class assignments for K=6:

# Store results in a data frame and name the variables:
consensusclass_p <- as.data.frame(results_p[[6]][["consensusClass"]])
names(consensusclass_p) <- "cluster"
# Set cluster as a factor and display a table:
consensusclass_p$cluster <- as.factor(consensusclass_p$cluster)
kable(table(consensusclass_p$cluster), col.names = c("Cluster", "Count"), format = "pipe" )
```

Cluster	Count
1	17
2	11
3	15
4	18
5	29
6	11

Retrieve features per cluster

```
# Create DataFrame with feature and group:
features_clust=consensusclass_p
# create a variable for features:
features_clust$feature <- rownames(features_clust)
# put "feature" first in data frame:
features_clust <- features_clust %>%
  select(feature, everything())
# set "cluster" as a factor:
features_clust$cluster = as.factor(features_clust$cluster)
# delete rownames:
rownames(features_clust) <- NULL

features_clust
```

feature <chr>	cluster <fct>
Elongation	1
Flatness	2
LeastAxisLength	2
MajorAxisLength	2
Maximum2DDiameterColumn	2
Maximum2DDiameterRow	2
Maximum2DDiameterSlice	2

feature <chr>	cluster <fct>
Maximum3DDiameter	2
MeshVolume	2
MinorAxisLength	2
1-10 of 101 rows	
Previous 1 2 3 4 5 6 ... 11 Next	

```
# Save results in an 'xls' file:
write.table(features_clust, file="./Data/features_clust.txt", quote=FALSE, sep="\t")
```

Reclassification Robustness

```
# With ClusterConsensus, calculate "cluster consensus" and "item consensus":
icl_p = calcICL(results_p, title="./Figures/Radiomic", plot="png")

# Store results for all k-values:
icl_clust_p <- data.frame(icl_p$clusterConsensus)

# Select and display results for k=6:
kable((k6_p <- round(icl_clust_p[icl_clust_p$k == 6,], 2)),
      caption = "Cluster consensus for k=6", format = "pipe")
```

Cluster consensus for k=6

	k	cluster	clusterConsensus
15	6	1	0.55
16	6	2	0.93
17	6	3	0.90
18	6	4	0.94
19	6	5	0.97
20	6	6	0.93

compute Average Patient Scores (APS) and map to clinical data

```

baselinedemo <- data.frame(
  id=rdr_L1_final$mask_id,
  D_SUBJID=rdr_L1_final$D_SUBJID,
  sex=rdr_L1_final$SEX.t1,
  age=rdr_L1_final$AGE,
  bmi=rdr_L1_final$BMI.t1,
  centre=rdr_L1_final$CENTREID,
  pksm = rdr_L1_final$SUPKYR,
  height = rdr_L1_final$HEIGHT,
  smokingstatus = rdr_L1_final$SMOKER)

emphysemaquart <- factor(cut(rdr_L1_final$FV950.t1, quantile(rdr_L1_final$FV950.t1, na.rm=TRUE)), labels = c(1,2,3,4))

emphysema <- factor(rdr_L1_final$FV950.t1>10, labels = c("No","Yes"))

exc <- as.numeric(rdr_L1_final$TOTEXBS)
exc[exc>3] <- 3

# Select GOLDCD and mask_id from ColData:
disease <- data.frame(
  goldcd=rdr_L1_final$GOLDCD,
  fve1=rdr_L1_final$FEV1PSDS.t1*1000,
  emphysema = emphysema,
  emphysemaquart=emphysemaquart,
  cbronch=rdr_L1_final$CBRONCH,
  fve2=rdr_L1_final$FEV1PSDS.t3*1000,
  fevdiffrdr_L1_final$FEV1PSDS.t3*1000 - rdr_L1_final$FEV1PSDS.t1*1000,
  exc=as.factor(exc)
)

clinnicaldat <- data.frame(baselinedemo, disease)

# access to assay data:
rad_data <- data.frame(assay(rdr_L1_final, "adjusted_min_int_center"))

# assign cluster to each corresponding feature:
rad_data <- cbind(rad_data, features_clust$cluster)

# set names
names(rad_data)[1774] <- "cluster"
feature_name <- rownames(rad_data)
rownames(rad_data) <- NULL
rad_data$feature <- feature_name
# Change colnames to match cov_data
colnames(rad_data) <- sub("X", "", colnames(rad_data))

# group by cluster. Compute the mean for each cluster:
rad_data <- aggregate(rad_data[,1:1773], by = list(rad_data$cluster), FUN = mean)
names(rad_data)[names(rad_data) == "Group.1"] <- "cluster"

```

```
# transpose radiomic data to have cluster means as columns:
```

```
rad_data <- data.frame(t(rad_data))
```

```
rad_data <- rad_data [-1,]
```

```
nms <- c("C1", "C2", "C3", "C4", "C5", "C6")
```

```
colnames(rad_data) <- nms
```

```
# Visualize both dataframes:
```

```
head(clinnicaldat)[,3:8]
```

	sex <chr>	age <int>	bmi <dbl>	centre <int>	pksm <int>	height <int>
1	F	70	26.85707	23140	60	157
2	F	57	31.67316	23140	26	161
3	F	51	22.37708	23140	12	152
4	M	55	35.38349	23140	37	185
5	M	63	16.13061	23140	50	175
6	M	66	36.10120	23140	30	174

6 rows

```
r_data <- rad_data
```

```
rownames(r_data) <- c()
```

```
head(r_data)[,1:6]
```

	C1 <chr>	C2 <chr>	C3 <chr>	C4 <chr>	C5 <chr>	C6 <chr>
1	0.68291091	-1.03979166	-0.06725801	-0.07463867	-0.02299889	0.05086614
2	0.001487651	-0.062449499	-0.132122541	0.368564749	-0.372587316	0.122834103
3	0.01957214	-1.20312898	0.09137449	-0.08179034	-0.06705381	0.05173055
4	-0.31492785	0.06766537	0.29636052	0.17392370	-0.24765629	0.11771286
5	0.63154785	0.77875999	-0.24363882	0.09734180	-0.13565503	0.08534437
6	-0.26863703	0.80919841	-0.07280032	0.44181654	-0.29470789	0.17037994

6 rows

```

# Merge three dataframes by subjects (row.names)
rad_data$Row.names <- rownames(rad_data)

model_data <- merge(clinnicaldat, rad_data,
                    by.x = 'id', by.y = "Row.names", all = TRUE)

rownames(model_data) <- model_data$id

# Set the internal structure of the dataset:
model_data <- model_data %>%
  mutate_at(c("id", "centre"), as.character) %>%
  mutate_at(c("sex",
              "goldcd", "cbronch", "smokingstatus", "emphysema", "emphysemaquart", "exc"), a
s.factor) %>%
  mutate_at(c("C1", "C2", "C3", "C4", "C5", "C6", "fve1", "fve2", "fevdifff", "age", "bmi", "p
ksm", "height"),
            as.numeric)

model_data <- model_data %>% mutate_at(c("C1", "C2", "C3", "C4", "C5", "C6"), scale)

save(model_data, file =  "./Data/model_data.rda")

write.table(model_data, file =  "./Data/model_data.txt", row=FALSE, quote=FALSE, sep="\t")

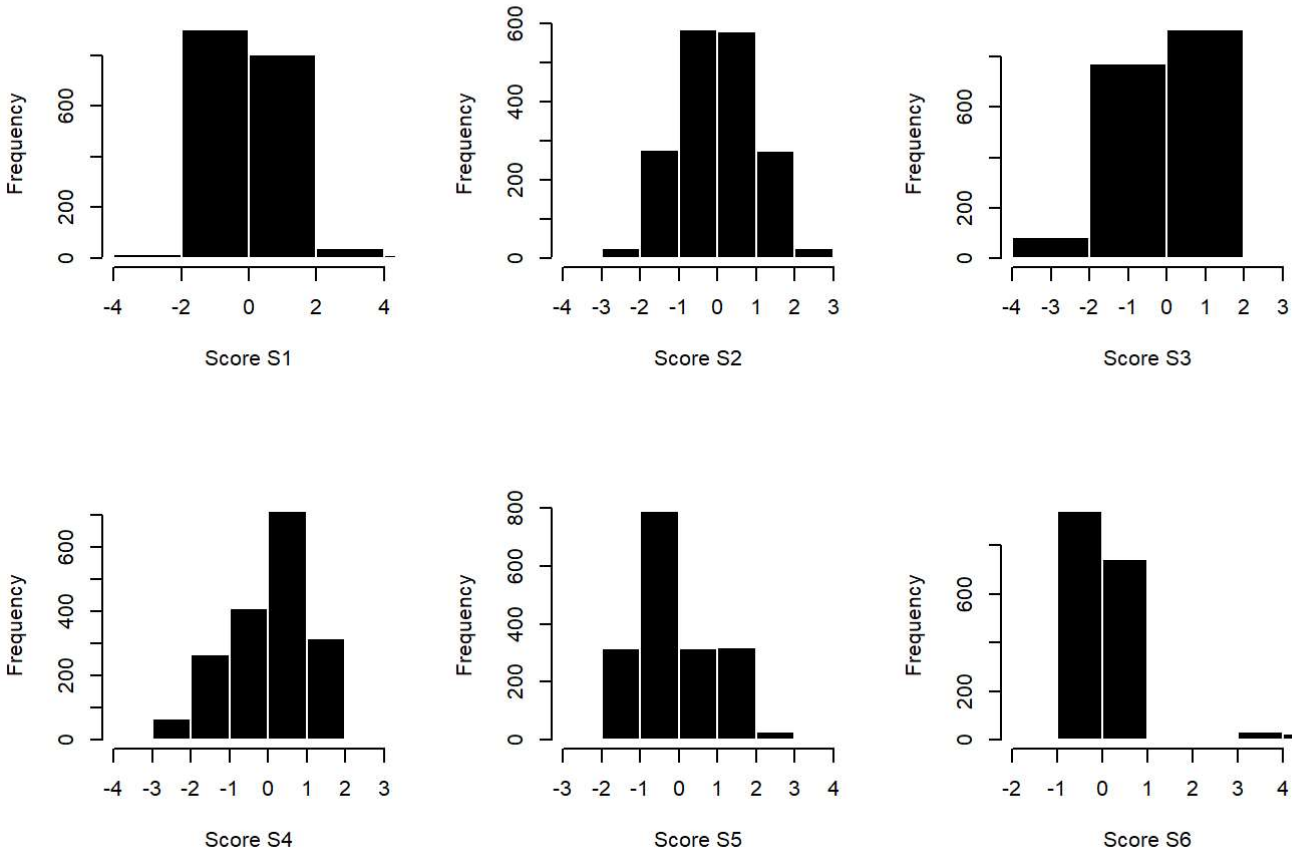
```

Distribution of APS

```

#png("hists.png", width = 2000, height = 1500, res = 300)
par(mfrow=c(2,3))
hist(model_data$C1, col="black", border = "white", xlab="Score S1", main=" ", br=6, xlim=c(-4,
4))
hist(model_data$C2, col="black", border = "white", xlab="Score S2", main=" ", br=6, xlim=c(-4,
3))
hist(model_data$C3, col="black", border = "white", xlab="Score S3", main=" ", br=6, xlim=c(-4,
3))
hist(model_data$C4, col="black", border = "white", xlab="Score S4", main=" ", br=6, xlim=c(-4,
3))
hist(model_data$C5, col="black", border = "white", xlab="Score S5", main=" ", br=6, xlim=c(-3,
4))
hist(model_data$C6, col="black", border = "white", xlab="Score S6", main=" ", br=6, xlim=c(-2,
4))

```



```
#dev.off()
```

Association of APSC(1-6) with baseline covariates

```
tabbase <- lapply(selmod, function(x){  
  
rr <- round(x[names(base_data),1:2], 4)  
  
out <- data.frame(paste(rr[,1], rr[,2], sep="±"),  
                  round(x[,3], 3))  
  
names(out) <- c("beta (± se)", "pval")  
rownames(out) <- names(base_data)  
out})  
  
write.table(t(do.call(cbind, tabbase)), file="tabbase.txt", sep="\t", quote=FALSE, row=TRUE,  
col=TRUE)  
  
do.call(cbind, tabbase)
```

	beta (± se) <chr>	pval <dbl>	beta (± se) <chr>	pval <dbl>	beta (± se) <chr>	pval <dbl>	beta (± se) <chr>
sex	0.0066±0.0572	0.908	0.7341±0.0433	0.000	-0.0218±0.0481	0.651	0.1824
age	-0.0078±0.0029	0.007	0.0018±0.0022	0.418	0.0048±0.0033	0.153	-0.0025

	beta (± se) <chr>	pval <dbl>	beta (± se) <chr>	pval <dbl>	beta (± se) <chr>	pval <dbl>	beta (± se) <chr>	pval <dbl>
bmi	-0.0877±0.0036	0.000	-0.0172±0.0027	0.000	0.0433±0.0041	0.000	-0.0783	
pksm	0.0011±8e-04	0.156	0.0013±6e-04	0.020	-8e-04±8e-04	0.376	-0.0025	
height	-0.0056±0.0023	0.014	0.0516±0.0023	0.000	-4e-04±0.0026	0.869	0.0018	
smokingstatus	-0.0183±0.0444	0.680	-0.0129±0.0328	0.693	0.1286±0.0488	0.008	0.1986	

Association of APSC(1-6) with baseline clinical variables

[1] 1

[1] 2

[1] 3

[1] 4

[1] 5

[1] 6

```
tabclin <- lapply(selmod, function(x){  
  
  rr <- round(x[,1:2], 2)  
  
  out <- data.frame(paste(rr[,1], rr[,2], sep="±"),  
                    round(x[,3], 3))  
  
  names(out) <- c("beta (± se)", "pval")  
  rownames(out) <- rownames(x)  
  out})  
  
write.table(do.call(cbind, tabclin), file="tabclin.txt", sep="\t", quote=FALSE, row=TRUE, col  
=TRUE)  
  
do.call(cbind, tabclin)
```

	beta (± se) <chr>	pval <dbl>	beta (± se) <chr>	pval <dbl>	beta (± se) <chr>	pval <dbl>	beta (± se) <chr>	pval <dbl>	beta (± se) <chr>	pval <dbl>
4>2	1.57±1.09	0.000	2.54±1.14	0.000	0.67±1.08	0.000	0.89±1.1	0.198	1.1	
4>3	1.29±1.1	0.005	1.58±1.14	0.000	0.85±1.08	0.028	0.97±1.09	0.762	1.0	
fve1	-98.74±12.43	0.000	-178.23±16.38	0.000	69.23±12.64	0.000	15.74±13.43	0.241	-24	
Yes>No	2.18±1.09	0.000	5.41±1.12	0.000	0.54±1.07	0.000	0.9±1.07	0.100	1.2	
Y>N	1.04±1.06	0.528	1.21±1.08	0.017	0.89±1.05	0.025	0.95±1.06	0.444	1.0	
3>0	1.24±1.08	0.003	1.56±1.1	0.000	0.82±1.07	0.001	0.84±1.07	0.013	1.2	
3>1	1.08±1.08	0.360	1.43±1.11	0.001	0.91±1.08	0.207	0.95±1.08	0.475	1.0	

	beta (± se) <chr>	pval <dbl>	beta (± se) <chr>	pval <dbl>	beta (± se) <chr>	pval <dbl>	beta (± se) <chr>	pval <dbl>	beta (± se) <chr>	pval <dbl>
3>2	0.96±1.09	0.622	1.11±1.12	0.357	1.05±1.08	0.555	0.93±1.09	0.383	1.0	

Association of APSC(1-6) with emphysema and GOLD grade

```
clusmeans <-unlist(model_data[,c("C1","C2","C3","C4","C5","C6")])
clus <- rep(c("C1","C2","C3","C4","C5","C6"), each=nrow(model_data))

dat <- lapply(1:6, function(x) model_data)
dat <- do.call(rbind, dat)
dat$clusmeans <- clusmeans
dat$clus <- factor(clus)

dat$clus <- factor(dat$clus, levels = rev(levels(dat$clus)))

dat <- na.omit(dat)

dd <- dat
rownames(dd) <- c()
head(dd[, 3:8])
```

	sex <fct>	age <dbl>	bmi <dbl>	centre <chr>	pksm <dbl>	height <dbl>
1	M	55	35.38349	23140	37	185
2	M	66	36.10120	23140	30	174
3	M	68	25.62732	23140	16	173
4	M	72	26.18339	23140	45	168
5	M	70	26.21882	23146	57	168
6	F	60	22.18773	23146	35	148
6 rows						

```

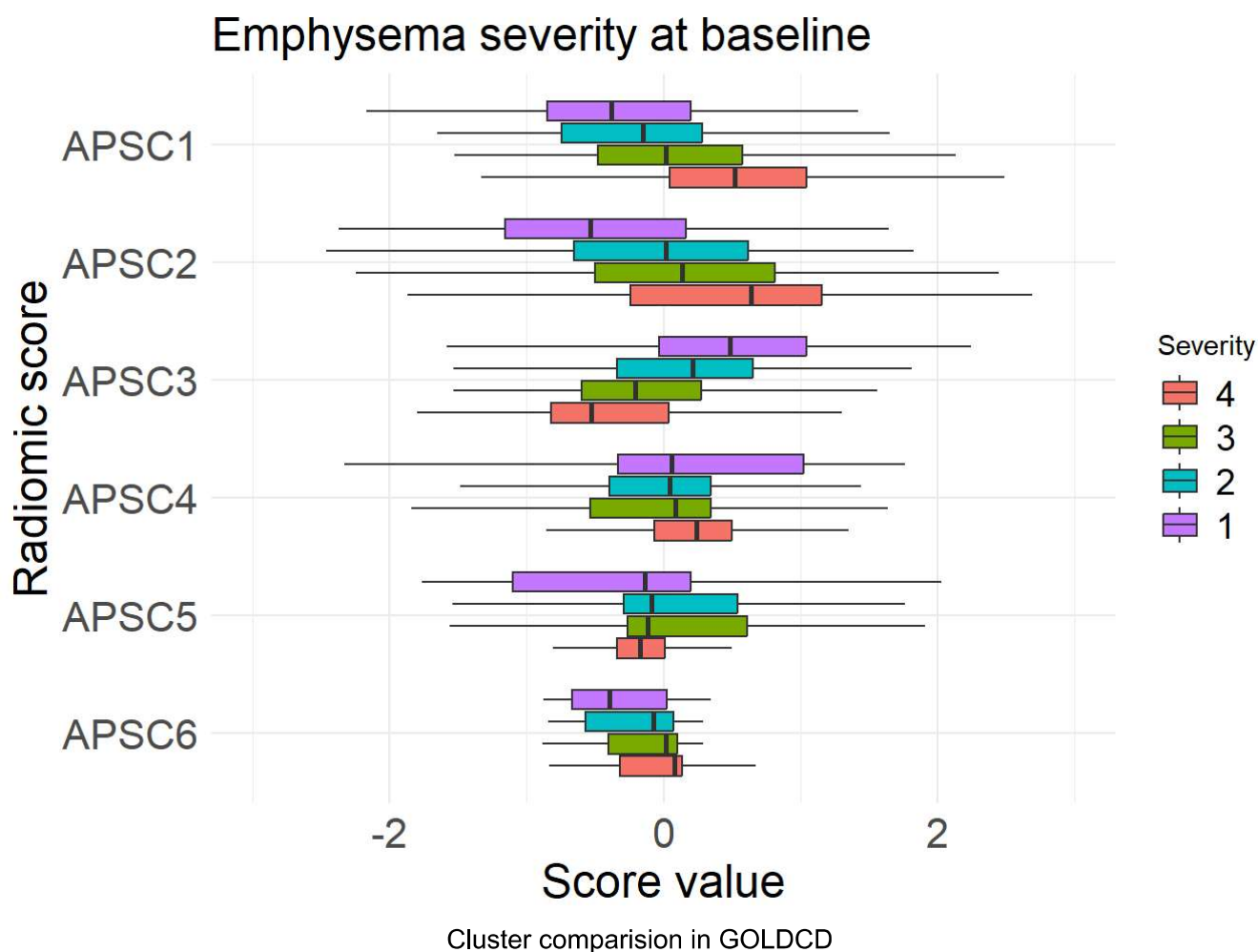
dat$emphysemaquart <- factor(dat$emphysemaquart,
                             levels=c("4", "3", "2", "1"))

pl <- ggplot(dat, aes(x=factor(clus, labels=c("APSC6", "APSC5", "APSC4", "APSC3", "APSC2", "APSC1")), y=clusmeans, fill=emphysemaquart)) +
  geom_boxplot(outlier.shape = NA) +
  ggtitle("Emphysema severity at baseline") +
  ylim(-3, 3) +
  labs(
    x = "Radiomic score",
    y = "Score value",
    fill = "Severity"
  ) +
  theme_minimal() +
  theme(
    axis.text = element_text(size = 16),
    axis.title = element_text(size = 18),
    plot.title = element_text(size = 18),
    legend.text = element_text(size = 14)
  ) +
  coord_flip()

#png("emph.png", width = 2000, height = 1500, res = 300)
pl

```

Warning: Removed 84 rows containing non-finite values (`stat\_boxplot()`).



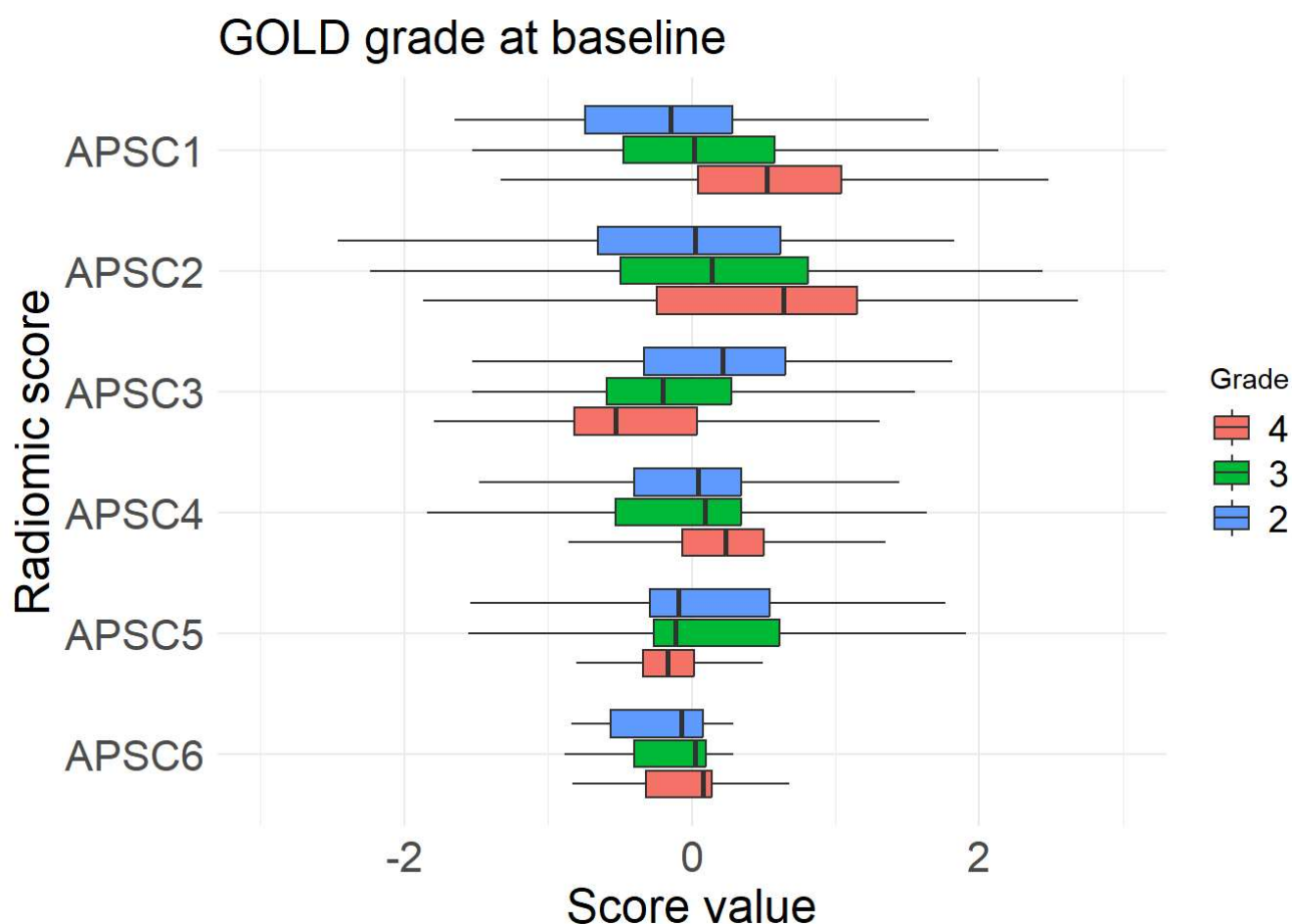
```
#dev.off()
```

```
dat$goldcd <- factor(dat$emphysemaquant,
                     levels=c("4", "3", "2"))

p1 <- ggplot(na.omit(dat), aes(x=factor(clus, labels=c("APSC6", "APSC5", "APSC4", "APSC3", "APSC2", "APSC1")), y=clusmeans, fill=goldcd)) +
  geom_boxplot(outlier.shape = NA) + ggtitle("GOLD grade at baseline ") + ylim(-3, 3) + labs(
    x = "Radiomic score",
    y = "Score value",
    fill = "Grade"
  ) +
  theme_minimal() +
  theme(
    axis.text = element_text(size = 16),
    axis.title = element_text(size = 18),
    plot.title = element_text(size = 18),
    legend.text = element_text(size = 14)
  ) +
  coord_flip()

#png("gold.png", width = 2000, height = 1500, res = 300)
p1
```

Warning: Removed 62 rows containing non-finite values (`stat\_boxplot()`).



```
#dev.off()
```

Associations of APSC(1-6) with survival

```
library(survival)  
library(survminer)
```

Loading required package: ggpubr

Attaching package: 'ggpubr'

The following object is masked from 'package:cowplot':

```
get_legend
```

Attaching package: 'survminer'

The following object is masked from 'package:survival':

```
myeloma
```

```
ids <- colData(rdr_L1_final)
ids <- ids[, c("id", "SUBJECT_ID")]
rownames(ids) <- ids$id

model_data <- merge(clinnicaldat, rad_data,
                    by.x = 'id', by.y = "Row.names", all = TRUE)

rownames(model_data) <- model_data$id

model_data <- model_data %>% mutate_at(c("C1", "C2", "C3", "C4", "C5", "C6"), as.numeric)

model_data <- cbind(model_data,
                    c1=model_data$C1,
                    c2=model_data$C2,
                    c3=model_data$C3,
                    c4=model_data$C4,
                    c5=model_data$C5,
                    c6=model_data$C6)

model_data <- model_data %>%
  mutate_at(c("C1", "C2", "C3", "C4", "C5", "C6"), scale)

model_data$SUBJECT_ID <- ids[rownames(model_data), "SUBJECT_ID"]

rownames(model_data) <- model_data$SUBJECT_ID

dat.surv <- read.table("./data/phs001252.v1.pht006380.v1.p1.c1.death.DS-COPD-RD.txt.gz", sep
                      = "\t", header=TRUE)

rownames(dat.surv) <- dat.surv$SUBJECT_ID

model_data.surv <- model_data

model_data.surv$months <- dat.surv[rownames(model_data),]$SURVTIME
model_data.surv$event <- as.numeric(factor(dat.surv[rownames(model_data),]$DIE3YR)=="Y")

mod1 <- coxph(Surv(months, event) ~ C1 + age + bmi + height, data=model_data.surv, x=TRUE)
summary(mod1)
```

Call:

```
coxph(formula = Surv(months, event) ~ C1 + age + bmi + height,
      data = model_data.surv, x = TRUE)
```

n= 1759, number of events= 158

(14 observations deleted due to missingness)

	coef	exp(coef)	se(coef)	z	Pr(> z)
C1	0.019467	1.019658	0.091363	0.213	0.8313
age	0.060968	1.062865	0.012641	4.823	1.42e-06 ***
bmi	-0.024292	0.976001	0.017385	-1.397	0.1623
height	-0.018605	0.981567	0.008779	-2.119	0.0341 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
C1	1.0197	0.9807	0.8525	1.2196
age	1.0629	0.9409	1.0369	1.0895
bmi	0.9760	1.0246	0.9433	1.0098
height	0.9816	1.0188	0.9648	0.9986

Concordance= 0.637 (se = 0.022)

Likelihood ratio test= 33.92 on 4 df, p=8e-07

Wald test = 31.4 on 4 df, p=3e-06

Score (logrank) test = 31.67 on 4 df, p=2e-06

```
mod2 <- coxph(Surv(months, event) ~ C2 + sex + bmi + height + pksm , data=model_data.surv, x=T
RUE)
summary(mod2)
```

Call:

```
coxph(formula = Surv(months, event) ~ C2 + sex + bmi + height +
      pksm, data = model_data.surv, x = TRUE)
```

n= 1759, number of events= 158

(14 observations deleted due to missingness)

	coef	exp(coef)	se(coef)	z	Pr(> z)
C2	0.323984	1.382625	0.132541	2.444	0.0145 *
sexM	0.368829	1.446040	0.247962	1.487	0.1369
bmi	-0.023819	0.976463	0.015112	-1.576	0.1150
height	-0.061876	0.940000	0.014073	-4.397	1.1e-05 ***
pksm	0.005804	1.005821	0.002567	2.262	0.0237 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
C2	1.3826	0.7233	1.0663	1.7928
sexM	1.4460	0.6915	0.8894	2.3510
bmi	0.9765	1.0241	0.9480	1.0058
height	0.9400	1.0638	0.9144	0.9663
pksm	1.0058	0.9942	1.0008	1.0109

Concordance= 0.623 (se = 0.021)

Likelihood ratio test= 30.3 on 5 df, p=1e-05

Wald test = 30.12 on 5 df, p=1e-05

Score (logrank) test = 30.13 on 5 df, p=1e-05

```
mod3 <- coxph(Surv(months, event) ~ C3 + bmi + smokingstatus , data=model_data.surv, x=TRUE)
summary(mod3)
```

Call:

```
coxph(formula = Surv(months, event) ~ C3 + bmi + smokingstatus,
      data = model_data.surv, x = TRUE)
```

n= 1759, number of events= 158

(14 observations deleted due to missingness)

	coef	exp(coef)	se(coef)	z	Pr(> z)
C3	-0.12353	0.88380	0.07828	-1.578	0.115
bmi	-0.02072	0.97950	0.01561	-1.327	0.184
smokingstatus	-0.08413	0.91931	0.16917	-0.497	0.619

	exp(coef)	exp(-coef)	lower .95	upper .95
C3	0.8838	1.131	0.7581	1.030
bmi	0.9795	1.021	0.9500	1.010
smokingstatus	0.9193	1.088	0.6599	1.281

Concordance= 0.544 (se = 0.024)

Likelihood ratio test= 5.98 on 3 df, p=0.1

Wald test = 6.18 on 3 df, p=0.1

Score (logrank) test = 6.14 on 3 df, p=0.1

```
mod4 <- coxph(Surv(months, event) ~ C4 + sex + bmi + height + smokingstatus + pksm , data=model_data.surv, x=TRUE)
summary(mod4)
```

Call:

```
coxph(formula = Surv(months, event) ~ C4 + sex + bmi + height +
      smokingstatus + pksm, data = model_data.surv, x = TRUE)
```

n= 1759, number of events= 158

(14 observations deleted due to missingness)

	coef	exp(coef)	se(coef)	z	Pr(> z)
C4	-0.026228	0.974113	0.092489	-0.284	0.776728
sexM	0.602915	1.827438	0.226668	2.660	0.007816 **
bmi	-0.033870	0.966698	0.017230	-1.966	0.049325 *
height	-0.043699	0.957242	0.011834	-3.693	0.000222 ***
smokingstatus	-0.081241	0.921972	0.170535	-0.476	0.633799
pksm	0.006098	1.006117	0.002520	2.420	0.015507 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
C4	0.9741	1.0266	0.8126	1.1677
sexM	1.8274	0.5472	1.1719	2.8496
bmi	0.9667	1.0344	0.9346	0.9999
height	0.9572	1.0447	0.9353	0.9797
smokingstatus	0.9220	1.0846	0.6600	1.2879
pksm	1.0061	0.9939	1.0012	1.0111

Concordance= 0.615 (se = 0.022)

Likelihood ratio test= 24.45 on 6 df, p=4e-04

Wald test = 24.91 on 6 df, p=4e-04

Score (logrank) test = 24.96 on 6 df, p=3e-04

```
mod5 <- coxph(Surv(months, event) ~ C5 + bmi + smokingstatus + pksm , data=model_data.surv,
x=TRUE)
summary(mod5)
```

Call:

```
coxph(formula = Surv(months, event) ~ C5 + bmi + smokingstatus +
      pksm, data = model_data.surv, x = TRUE)
```

n= 1759, number of events= 158

(14 observations deleted due to missingness)

	coef	exp(coef)	se(coef)	z	Pr(> z)
C5	0.042175	1.043077	0.091695	0.460	0.64555
bmi	-0.035817	0.964817	0.017312	-2.069	0.03855 *
smokingstatus	-0.103212	0.901936	0.170386	-0.606	0.54468
pksm	0.006807	1.006831	0.002465	2.762	0.00574 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
C5	1.0431	0.9587	0.8715	1.2484
bmi	0.9648	1.0365	0.9326	0.9981
smokingstatus	0.9019	1.1087	0.6459	1.2595
pksm	1.0068	0.9932	1.0020	1.0117

Concordance= 0.589 (se = 0.023)

Likelihood ratio test= 10.71 on 4 df, p=0.03

Wald test = 11.23 on 4 df, p=0.02

Score (logrank) test = 11.23 on 4 df, p=0.02

```
mod6 <- coxph(Surv(months, event) ~ C6 + bmi , data=model_data.surv, x=TRUE)
summary(mod6)
```

Call:

```
coxph(formula = Surv(months, event) ~ C6 + bmi, data = model_data.surv,
      x = TRUE)
```

n= 1759, number of events= 158

(14 observations deleted due to missingness)

	coef	exp(coef)	se(coef)	z	Pr(> z)
C6	0.10042	1.10564	0.06742	1.489	0.136
bmi	-0.01921	0.98098	0.01535	-1.251	0.211

	exp(coef)	exp(-coef)	lower .95	upper .95
C6	1.106	0.9045	0.9688	1.262
bmi	0.981	1.0194	0.9519	1.011

Concordance= 0.556 (se = 0.025)

Likelihood ratio test= 5.21 on 2 df, p=0.07

Wald test = 5.75 on 2 df, p=0.06

Score (logrank) test = 5.86 on 2 df, p=0.05

```

dat.pi10 <- read.table("./data/phs001252.v1.pht006402.v1.p1.c1.vdawal1.DS-COPD-RD.txt.gz", sep="\t", header=TRUE)

dat.pi10 <- dat.pi10[dat.pi10$VISIT=="Screening",]

dat.pi10 <- dat.pi10[!is.na(dat.pi10$AWPI10_B),]

rownames(dat.pi10) <- dat.pi10$SUBJECT_ID

model_data.pi10 <- model_data
model_data.pi10$pi10 <- dat.pi10[rownames(model_data),]$AWPI10

mod1 <- lm(pi10 ~ C1 + age + bmi + height, data=model_data.pi10)
summary(mod1)$coeff[2,]

```

Estimate	Std. Error	t value	Pr(> t)
0.0163178143	0.0047424004	3.4408343639	0.0005947702

```

mod2 <- lm(pi10 ~ C2 + sex + bmi + height + pksm , data=model_data.pi10)
summary(mod2)$coeff[2,]

```

Estimate	Std. Error	t value	Pr(> t)
0.008219545	0.005727957	1.434987399	0.151485395

```

mod3 <- lm(pi10 ~ C3 + bmi + smokingstatus , data=model_data.pi10)
summary(mod3)$coeff[2,]

```

Estimate	Std. Error	t value	Pr(> t)
-0.007554651	0.004432595	-1.704340342	0.088510707

```

mod4 <- lm(pi10 ~ C4 + sex + bmi + height + smokingstatus + pksm , data=model_data.pi10)
summary(mod4)$coeff[2,]

```

Estimate	Std. Error	t value	Pr(> t)
2.588853e-02	4.169571e-03	6.208918e+00	6.779406e-10

```

mod5 <- lm(pi10 ~ C5 + bmi + smokingstatus + pksm , data=model_data.pi10)
summary(mod5)$coeff[2,]

```

Estimate	Std. Error	t value	Pr(> t)
-3.175141e-02	4.900463e-03	-6.479266e+00	1.222277e-10

```

mod6 <- lm(pi10 ~ C6 + bmi , data=model_data.pi10)
summary(mod6)$coeff[2,]

```

Estimate	Std. Error	t value	Pr(> t)
0.013257020	0.004321229	3.067881816	0.002191785

Associations of APSC(1-6) with FEV rate of change

Format data

```
ids <- colData(rdr_L1_final)
ids <- ids[, c("id", "SUBJECT_ID")]
rownames(ids) <- ids$id

model_data <- merge(clinnicaldat, rad_data,
                    by.x = 'id', by.y = "Row.names", all = TRUE)

rownames(model_data) <- model_data$id

model_data <- model_data %>% mutate_at(c("C1", "C2", "C3", "C4", "C5", "C6"), as.numeric)

model_data <- cbind(model_data,
                    c1=model_data$C1,
                    c2=model_data$C2,
                    c3=model_data$C3,
                    c4=model_data$C4,
                    c5=model_data$C5,
                    c6=model_data$C6)

model_data <- model_data %>%
  mutate_at(c("C1","C2","C3","C4","C5","C6"), scale)

model_data$SUBJECT_ID <- ids[rownames(model_data), "SUBJECT_ID"]

rownames(model_data)<-model_data$SUBJECT_ID

dat.visits <- read.table("../data/phs001252.v1.pht006393.v1.p1.c1.pft.DS-COPD-RD.txt.gz", sep
="\\t", header=TRUE)

visit <- factor(dat.visits$VISIT, labels = c(1,1.5,2,2.5,0.25,3,0.5,NA, 0,NA))

dat.visits$FEV1PSDS <- dat.visits$FEV1PSDS *1000

fevtime <- lapply(sort(levels(visit)[-8]), function(x){

  out <- dat.visits[visit%in%x,]
  rownames(out) <- out$SUBJECT_ID
  out <- out[rownames(model_data), "FEV1PSDS", drop=FALSE]
  rownames(out) <- rownames(model_data)
  out
})

fevtime <- do.call(cbind, fevtime)

colnames(fevtime) <- sort(levels(visit)[-8])

model_dat_times <- data.frame(model_data, fevtime)

fve <-unlist(fevtime)
```

```
time <- rep(sort(levels(visit)[-8]), each=nrow(model_dat_times))

dat.long <- lapply(1:8, function(x) model_dat_times)
dat.long <- do.call(rbind, dat.long)
dat.long$fve <- fve
dat.long$time <- as.numeric(time)

dd <- dat.long
rownames(dd) <- c()

head(dd[,3:8])
```

	sex <chr>	age <int>	bmi <dbl>	centre <int>	pksm <int>	height <int>
1	F	70	26.85707	23140	60	157
2	F	57	31.67316	23140	26	161
3	F	51	22.37708	23140	12	152
4	M	55	35.38349	23140	37	185
5	M	63	16.13061	23140	50	175
6	M	66	36.10120	23140	30	174

6 rows

```
sellvars <- c("fve", "age", "sex", "bmi", "time",
             "exc", "cbronch", "smokingstatus", "pksm", "height",
             "emphysema",
             "C1", "C2", "C3", "C4", "C5", "C6",
             "c1", "c2", "c3", "c4", "c5", "c6",
             "id", "goldcd", "fevdiff")

dat.long <- dat.long[,sellvars]
```

FEV1 change by covariates

```
dat <- dat.long

#Base model from Vetsbo
model <- lmer(
  fve ~ age + sex + bmi* time +
    smokingstatus* time+ pksm + height + emphysema * time + (1 + time | id),
  data = dat
)

summary(model)
```



```

pksm
height
emphysemaYs
bmi:time
tm:smkngstt 0.226
tm:mphysmYs 0.160 0.264

```

```

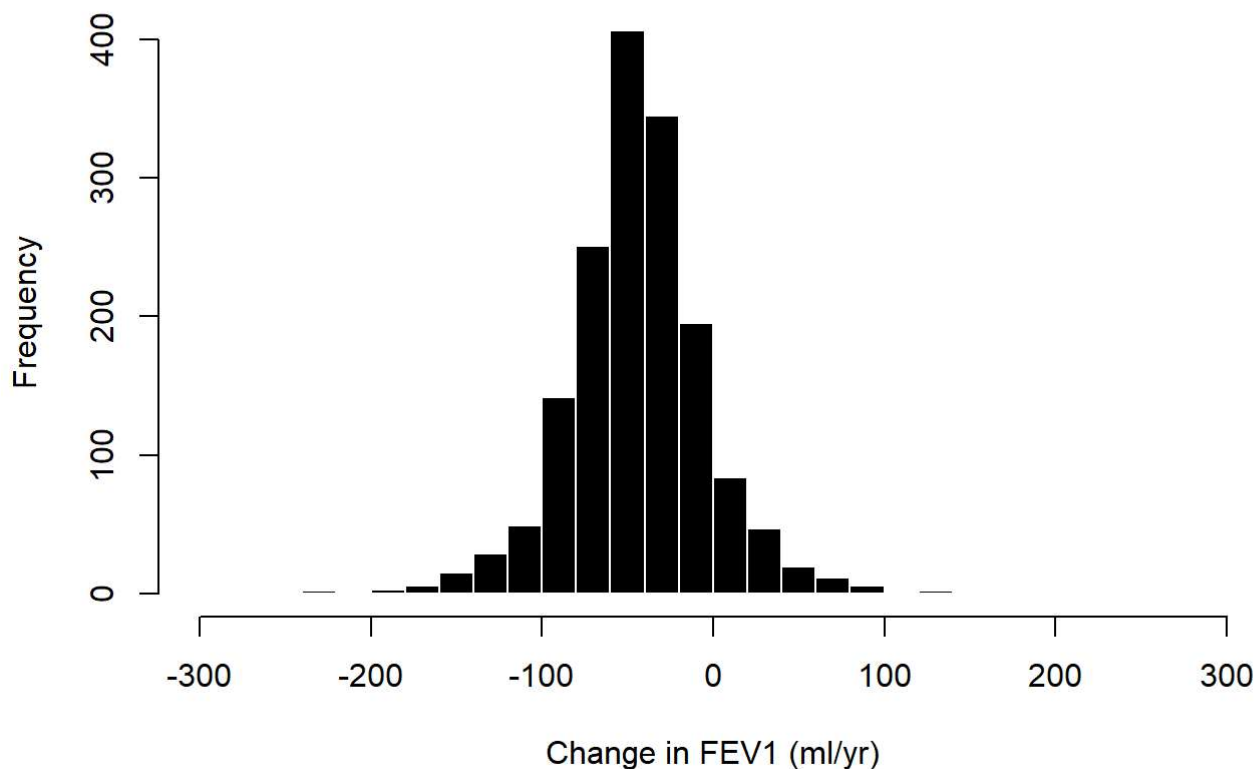
random_effects <- ranef(model)$id["time"]

ci <- confint(model)

#png("hist.png", width = 2000, height = 1500, res = 300)

hist(random_effects+fixef(model)["time"], freq=TRUE,
      xlim=c(-300, 300), xlab="Change in FEV1 (ml/yr)", br=20,
      main="", col="black", border = "white")

```



```
#dev.off()
```

FEV1 change by APSC(1-6)

```
clusnames <- c("C1", "C2", "C3", "C4", "C5", "C6")

selmod <- lapply(clusnames, function(cc){
  dd <- dat
  dd$clus <- dat[,cc]

  model <- lmer(fve~age + sex + bmi* time + exc + cbronch + smokingstatus* time + pksm + height
+ emphysema*time + clus*time + (1 + time | id), data = dd
)

  out <- summary(model)$coeff[c("clus","time:clus"),c(1,2,5)]

  rownames(out) <- paste(c("", "time:"), cc, sep="")
  ci <- confint(model)

  out <- cbind(out,ci[c("clus", "time:clus"),])

  out
})

tabclus <- lapply(selmod, function(x){

  rr <- round(x[,1:2], 2)

  out <- data.frame(paste(rr[,1], rr[,2], sep="±"),
                    round(x[,3], 3), round(x[,4:5],3))

  names(out) <- c("beta (± se)", "pval", "l", "u")

  out})

do.call(rbind, tabclus)
```

	beta (± se) <chr>	pval <dbl>	l <dbl>	u <dbl>
C1	-57.31±13.09	0.000	-82.878	-31.755
time:C1	-2.5±2.62	0.340	-7.623	2.630
C2	-82.3±17.64	0.000	-116.765	-47.825
time:C2	-5.26±2.19	0.016	-9.553	-0.975
C3	36.8±11.13	0.001	15.066	58.542
time:C3	5.24±2.25	0.020	0.822	9.649
C4	3.02±11.99	0.801	-20.399	26.429
time:C4	-0.91±2.39	0.704	-5.594	3.779

	beta (± se) <chr>	pval <dbl>	l <dbl>	u <dbl>
C5	-6.9±12.05	0.567	-30.437	16.644
time:C5	0.26±2.41	0.913	-4.465	4.990
1-10 of 12 rows			Previous	1 2 Next

```
write.table(do.call(rbind, tabclus), file="tabclus.txt", sep="\t", quote=FALSE, row=TRUE, col=TRUE)
```

FEV1 change by APSC(1-6) stratified by GOLD

```
##BY GOLD
```

```
tablgold<- lapply(c("2", "3", "4"), function(gc){
```

```
  selmod <- lapply(c("C1", "C2", "C3", "C4", "C5", "C6"), function(cc){
```

```
    dd <- dat
```

```
    dd$clus <- dat[,cc]
```

```
    model <- lmer(
```

```
      fve ~ age + sex + bmi* time + exc + cbronch+
```

```
        smokingstatus* time + pksm + height + emphysema * time +
```

```
        clus*time+(1 + time | id),
```

```
      data = dd, subset = which(dd$goldcd==gc)
```

```
    )
```

```
    out <- summary(model)$coeff[c("clus", "time:clus"),c(1,2,5)]
```

```
    rownames(out) <- paste(c("", "time:"), cc, sep="")
```

```
    ci <- confint(model)
```

```
    out <- cbind(out,ci[c("clus", "time:clus"),])
```

```
  out
```

```
})
```

```
tabbase <- lapply(selmod, function(x){
```

```
  rr <- round(x[,1:2], 2)
```

```
  out <- data.frame(paste(rr[,1], rr[,2], sep="±"),
```

```
                    round(x[,3], 3), round(x[,4:5],3))
```

```
  names(out) <- c("beta (± se)", "pval", "l", "u")
```

```
  out})
```

```
do.call(rbind, tabbase)
```

```
})
```

```
write.table(do.call(cbind, tablgold), file="tablgold.txt", sep="\t", quote=FALSE, row=TRUE, c
ol=TRUE)
```

Prediction optimization

```
dat <- dat.long
ccases<-complete.cases(dat[, -c(18,19)])
dat <- dat[ccases,]

res <- lm(
  fve ~ age + sex + bmi + exc + cbronch+
    smokingstatus+ pksm + height + emphysema,
  data = dat)

dat$res <- summary(res)$residuals
```

APSC2 optimization cut-off

```
th <- seq(-1.2, 1.2, by = 0.1)

opt <- sapply(1:length(th), function(x){
  dat$clus <- factor(dat$C2<th[x], labels = c("> q", "< q"))

  mod0 <- lmer(
    fve ~ age + sex + bmi* time + exc + cbronch+
      smokingstatus* time+ pksm + height+ emphysema * time + clus + (1 + time| id),
    data = dat)

  mod <- lmer(
    fve ~ age + sex + bmi* time + exc + cbronch+
      smokingstatus* time+ pksm + height+ emphysema * time +
      clus*time+(1 + time| id),
    data = dat)

  summary(mod)$coef["time:clus< q",5]

  anova(mod0,mod, test="LRT")$'Pr(>Chisq)'[2]

})

t <- th[which(opt[1:20]==min(opt[1:20]))]
```

```
dat$clus <- factor(dat$C2> -0.4, labels = c("low (< -0.4)", "high (> -0.4)"))

modres <- lmer(
  res ~ age + sex + bmi* time + exc + cbronch+
    smokingstatus* time+ pksm+ height+emphysema * time +
    clus*time+(1 + time| id),
  data = dat)

summary(modres)$coeff["time:clushigh (> -0.4)",]
```

Estimate	Std. Error	df	t value	Pr(> t)
-1.209306e+01	4.552147e+00	1.233314e+03	-2.656563e+00	7.996037e-03

```
confint(modres)
```

```
Computing profile confidence intervals ...
```

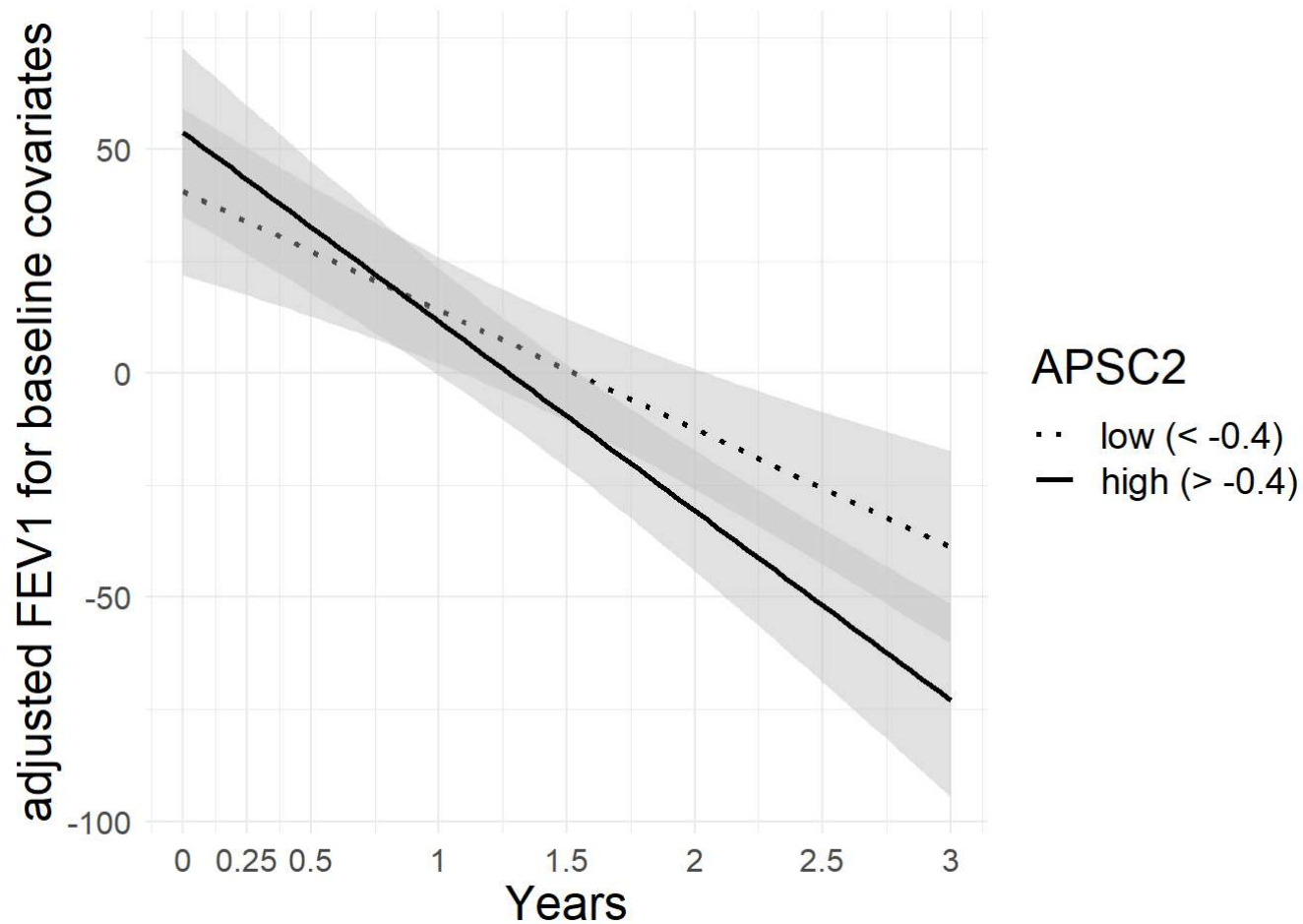
	2.5 %	97.5 %
.sig01	380.7884463	413.30782426
.sig02	-0.2141585	-0.07169834
.sig03	53.6598671	61.19541685
.sigma	130.1298654	134.44014258
(Intercept)	-624.2883998	666.36654776
age	-3.5002577	3.22743245
sexM	-62.8312674	73.31801598
bmi	-5.8638663	2.54468175
time	-71.7030919	-24.63589638
exc1	-70.5545656	73.87276230
exc2	-70.9227572	79.10378715
exc3	-59.4274710	58.53819727
cbronchY	-47.4185363	50.21592887
smokingstatus	-29.2400903	73.52542444
pksm	-0.8869615	0.83829317
height	-3.1485114	3.88516462
emphysemaYes	-42.4304023	63.41242306
clushigh (> -0.4)	-68.7890589	66.27446597
bmi:time	0.2957448	1.81141873
time:smokingstatus	-24.4252841	-6.25919418
time:emphysemaYes	-14.2510107	4.49392942
time:clushigh (> -0.4)	-21.0042954	-3.18241234

```
pred <- data.frame(dat, predicted=predict(modres))

plot1 <- ggplot(pred, aes(x = time, y = res, linetype = clus)) +
  geom_smooth(aes(fill = clus), method = "lm", se = TRUE, color = "black") +
  scale_linetype_manual(values = c("dotted", "solid"), name = "APSC2")+
  scale_fill_manual(values = c("grey", "grey"), guide = "none")+
  guides(
    linetype = guide_legend(
      override.aes = list(fill = "white")
    ) +
    scale_x_continuous(breaks = c(0.0, 0.25, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0)
      , labels = c("0", "0.25", "0.5", "1", "1.5", "2", "2.5", "3"))+
  labs(x = "Years", y = "adjusted FEV1 for baseline covariates") +
  theme_minimal()+
  theme(
    axis.text = element_text(size = 12),
    axis.title = element_text(size = 18),
    plot.title = element_text(size = 18),
    legend.text = element_text(size = 14),
    legend.title = element_text(size = 18))

#png("intC2Adjusted.png", width = 2000, height = 1500, res = 300)
plot1
```

```
`geom_smooth()` using formula = 'y ~ x'
```



```
#dev.off()
```

By GOLD

```

t1 <- t
t2 <- t
t3 <- t

dat$clus <- NA
dat$clus[dat$goldcd==2 & dat$C2 < t1] <- "low"
dat$clus[dat$goldcd==2 & dat$C2 > t1] <- "high"

dat$clus[dat$goldcd==3 & dat$C2 < t2] <- "low"
dat$clus[dat$goldcd==3 & dat$C2 > t2] <- "high"

dat$clus[dat$goldcd==4 & dat$C2 < t3] <- "low"
dat$clus[dat$goldcd==4 & dat$C2 > t3] <- "high"

dat$clus <- factor(dat$clus, levels = c("low", "high"))

out <- c()
for(x in c(2,3,4)){
  mod <- lmer(
    fve ~ age + sex + bmi* time + exc + cbronch+
      smokingstatus* time+ pksm+ height+emphysema * time +
      clus*time+(1 + time| id),
    data = dat, subset = which(dat$goldcd==x))

  out <- rbind(out, summary(mod)$coeff["time:clushigh",])
}

out

```

	Estimate	Std. Error	df	t value	Pr(> t)
[1,]	-10.188364	6.875146	585.0825	-1.4819123	0.138901984
[2,]	-21.789200	6.926382	499.7031	-3.1458269	0.001754898
[3,]	-1.317207	10.904477	137.6897	-0.1207951	0.904029441

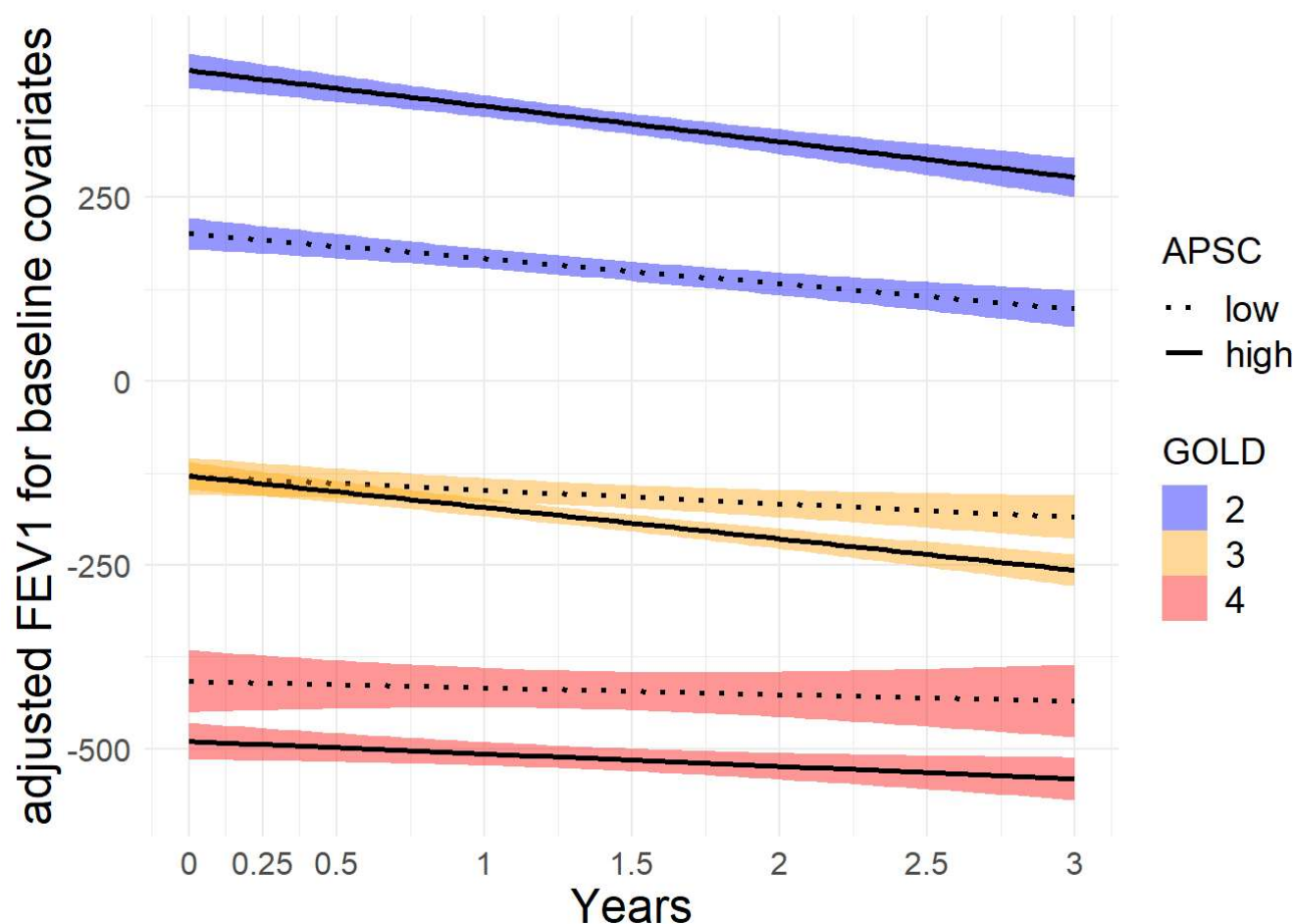
```

pred <- data.frame(dat, predicted=predict(modres))

plot1 <- ggplot(pred, aes(x = time, y = res, linetype = factor(clus))) +
  geom_smooth(aes(fill = factor(goldcd)), method = "lm", se = TRUE, color = "black") +
  scale_linetype_manual(values = c("dotted", "solid"), name = "APSC")+
  scale_fill_manual(values = c("blue", "orange", "red"), name = "GOLD")+
  guides(
    linetype = guide_legend(
      override.aes = list(fill = "white")),
    fill = guide_legend(
      override.aes = list(linetype = "blank") # Add Line type to GOLD Legend
    )
  )+
  labs(x = "Years", y = "adjusted FEV1 for baseline covariates") +
  scale_x_continuous(breaks = c(0.0, 0.25, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0)
    , labels = c("0", "0.25", "0.5", "1", "1.5", "2", "2.5", "3")) +
  theme_minimal() +
  theme(legend.title = element_text(size = 14),
    axis.text = element_text(size = 12),
    axis.title = element_text(size = 18),
    plot.title = element_text(size = 18),
    legend.text = element_text(size = 14))
#png("intC2AdjustedByGOLD.png", width = 2000, height = 1500, res = 300)
plot1

```

```
`geom_smooth()` using formula = 'y ~ x'
```



```
#dev.off()
```

APSC3 optimization cut-off

```
th <- seq(-1.2, 1.2, by = 0.1)

opt <- sapply(1:length(th), function(x){
  dat$clus <- factor(dat$C3<th[x], labels = c("> q", "< q"))

  mod0 <- lmer(
    fve ~ age + sex + bmi* time + exc + cbronch+
      smokingstatus* time+ pksm + height+ emphysema * time + clus + (1 + time| id),
    data = dat)

  mod <- lmer(
    fve ~ age + sex + bmi* time + exc + cbronch+
      smokingstatus* time+ pksm + height+ emphysema * time +
      clus*time+(1 + time| id),
    data = dat)

  summary(mod)$coef["time:clus< q",5]

  anova(mod0,mod, test="LRT")$'Pr(>Chisq)')[2]

})

t <- th[which(opt==min(opt))]
```

```
dat$clus <- factor(dat$C3> -0.6, labels = c("low (< -0.6)", "high (> -0.6)"))

modres <- lmer(
  res ~ age + sex + bmi* time + exc + cbronch+
    smokingstatus* time+ pksm+ height+emphysema * time +
    clus*time+(1 + time| id),
  data = dat)

summary(modres)$coeff["time:clushigh (> -0.6)",]
```

Estimate	Std. Error	df	t value	Pr(> t)
13.8603014	5.3445079	1233.1578665	2.5933728	0.0096165

```
confint(modres)
```

```
Computing profile confidence intervals ...
```

	2.5 %	97.5 %
.sig01	377.11260142	409.34784800
.sig02	-0.22689984	-0.08494301
.sig03	53.67201429	61.20747494
.sigma	130.12941782	134.43964667
(Intercept)	-625.46101830	602.71284971
age	-3.97872732	2.67827741
sexM	-63.93057163	59.67489865
bmi	-8.58048968	0.03065389
time	-81.66873795	-34.30378577
exc1	-59.20663813	83.73791139
exc2	-57.70944090	90.76295662
exc3	-51.18507516	65.53692335
cbronchY	-40.06206562	56.53703007
smokingstatus	-30.63976284	70.78076731
pksm	-0.77833254	0.92828376
height	-2.89793193	3.65435232
emphysemaYes	-17.05872011	84.65385013
clushigh (> -0.6)	81.70453873	196.38354597
bmi:time	-0.01242518	1.55232291
time:smokingstatus	-25.44052138	-7.34318250
time:emphysemaYes	-15.23460640	3.11919190
time:clushigh (> -0.6)	3.39799579	24.32202381

```

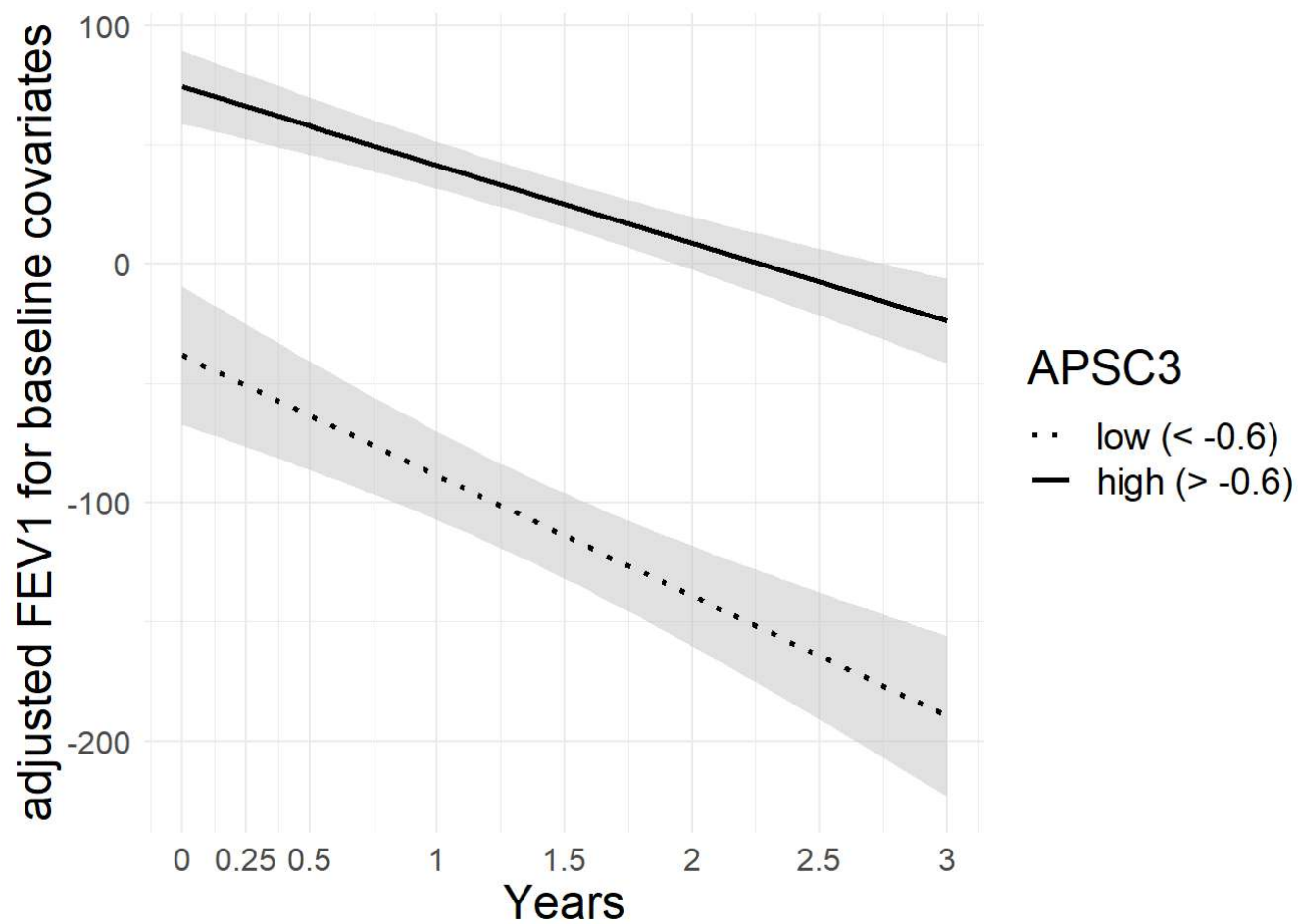
pred <- data.frame(dat, predicted=predict(modres))

plot1 <- ggplot(pred, aes(x = time, y = res, linetype = clus)) +
  geom_smooth(aes(fill = clus), method = "lm", se = TRUE, color = "black") +
  scale_linetype_manual(values = c("dotted", "solid"), name = "APSC3")+
  scale_fill_manual(values = c("grey", "grey"), guide = "none")+
  guides(
    linetype = guide_legend(
      override.aes = list(fill = "white"))
  ) +
  scale_x_continuous(breaks = c(0.0, 0.25, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0)
    , labels = c("0", "0.25", "0.5", "1", "1.5", "2", "2.5", "3"))+
  labs(x = "Years", y = "adjusted FEV1 for baseline covariates") +
  theme_minimal()+
  theme(
    axis.text = element_text(size = 12),
    axis.title = element_text(size = 18),
    plot.title = element_text(size = 18),
    legend.text = element_text(size = 14),
    legend.title = element_text(size = 18))

#png("intC3Adjusted.png", width = 2000, height = 1500, res = 300)
plot1

```

```
`geom_smooth()` using formula = 'y ~ x'
```



```
#dev.off()
```

By GOLD

```

t1 <- t
t2 <- t
t3 <- t

dat$clus <- NA
dat$clus[dat$goldcd==2 & dat$C3 < t1] <- "low"
dat$clus[dat$goldcd==2 & dat$C3 > t1] <- "high"

dat$clus[dat$goldcd==3 & dat$C3 < t2] <- "low"
dat$clus[dat$goldcd==3 & dat$C3 > t2] <- "high"

dat$clus[dat$goldcd==4 & dat$C3 < t3] <- "low"
dat$clus[dat$goldcd==4 & dat$C3 > t3] <- "high"

dat$clus <- factor(dat$clus, levels = c("low", "high"))

out <- c()
for(x in c(2,3,4)){
  mod <- lmer(
    fve ~ age + sex + bmi* time + exc + cbronch+
      smokingstatus* time+ pksm+ height+emphysema * time +
      clus*time+(1 + time| id),
    data = dat, subset = which(dat$goldcd==x))

  out <- rbind(out, summary(mod)$coeff["time:clushigh",])
}

out

```

	Estimate	Std. Error	df	t value	Pr(> t)
[1,]	13.46313	9.791929	584.2928	1.374921	0.169682817
[2,]	20.66083	7.396456	498.7623	2.793342	0.005417239
[3,]	18.75366	9.451749	137.5113	1.984147	0.049230285

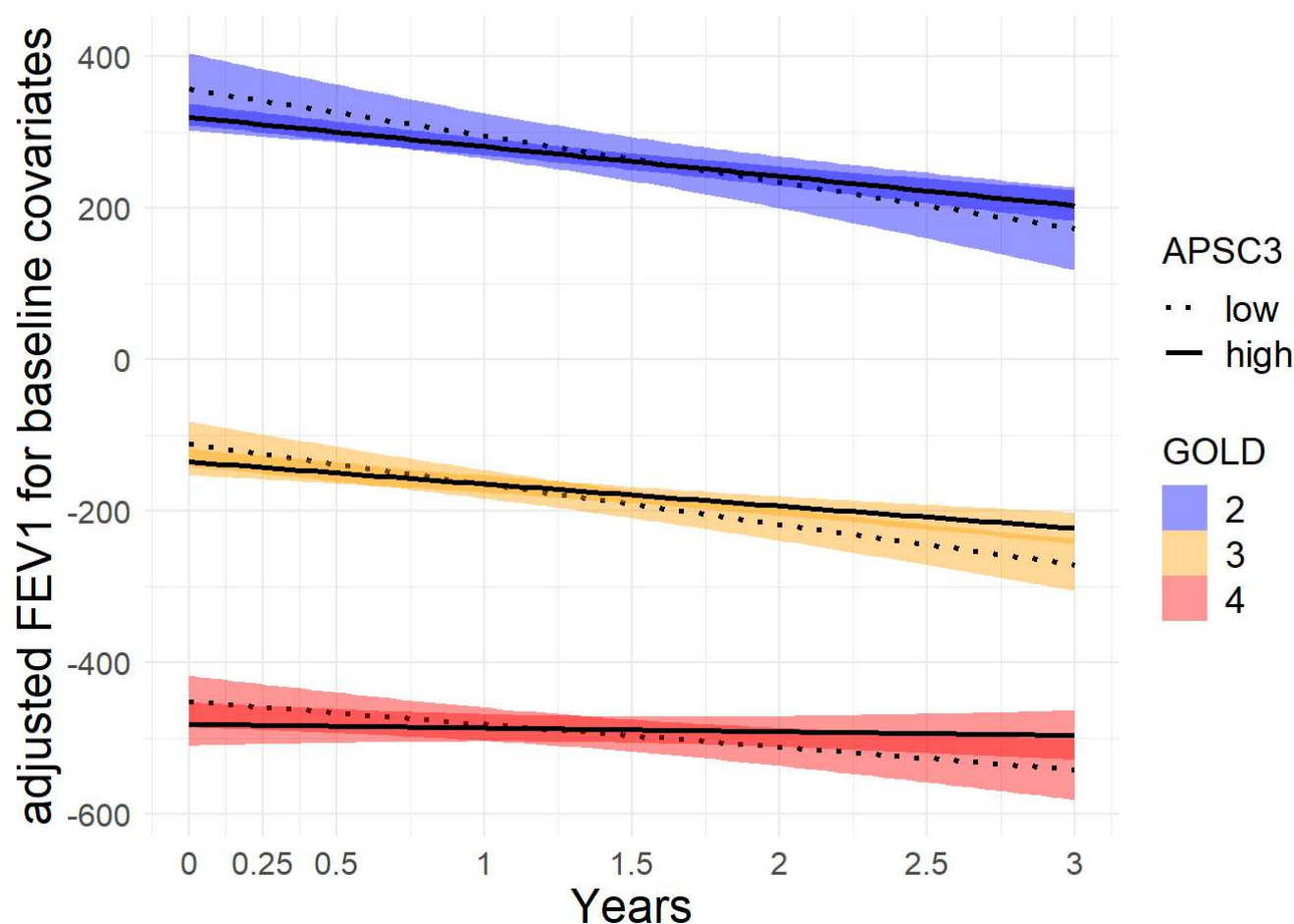
```

pred <- data.frame(dat, predicted=predict(modres))

plot1 <- ggplot(pred, aes(x = time, y = res, linetype = factor(clus))) +
  geom_smooth(aes(fill = factor(goldcd)), method = "lm", se = TRUE, color = "black") +
  scale_linetype_manual(values = c("dotted", "solid"), name = "APSC3")+
  scale_fill_manual(values = c("blue", "orange", "red"), name = "GOLD")+
  guides(
    linetype = guide_legend(
      override.aes = list(fill = "white")),
    fill = guide_legend(
      override.aes = list(linetype = "blank") # Add Line type to GOLD Legend
    )
  )+
  labs(x = "Years", y = "adjusted FEV1 for baseline covariates") +
  scale_x_continuous(breaks = c(0.0, 0.25, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0)
    , labels = c("0", "0.25", "0.5", "1", "1.5", "2", "2.5", "3")) +
  theme_minimal() +
  theme(legend.title = element_text(size = 14),
    axis.text = element_text(size = 12),
    axis.title = element_text(size = 18),
    plot.title = element_text(size = 18),
    legend.text = element_text(size = 14))
#png("intC3AdjustedByGOLD.png", width = 2000, height = 1500, res = 300)
plot1

```

```
`geom_smooth()` using formula = 'y ~ x'
```



```
#dev.off()
```

APSC6 optimization cut-off

```
th <- seq(-0.6, 0.6, by = 0.1)

opt <- sapply(1:length(th), function(x){
  dat$clus <- factor(dat$C6<th[x], labels = c("> q", "< q"))

  mod0 <- lmer(
    fve ~ age + sex + bmi* time + exc + cbronch+
      smokingstatus* time+ pksm + height+ emphysema * time + clus + (1 + time| id),
    data = dat)

  mod <- lmer(
    fve ~ age + sex + bmi* time + exc + cbronch+
      smokingstatus* time+ pksm + height+ emphysema * time +
      clus*time+(1 + time| id),
    data = dat)

  summary(mod)$coef["time:clus< q",5]

  anova(mod0,mod, test="LRT")$'Pr(>Chisq)'[2]

})

t <- th[which(opt==min(opt))]
```

```
dat$clus <- factor(dat$C6> 0.2, labels = c("low (< 0.2)", "high (> 0.2)"))

modres <- lmer(
  res ~ age + sex + bmi* time + exc + cbronch+
    smokingstatus* time+ pksm+ height+emphysema * time +
    clus*time+(1 + time| id),
  data = dat)

summary(modres)$coeff["time:clushigh (> 0.2)",]
```

Estimate	Std. Error	df	t value	Pr(> t)
-2.268358e+01	8.113264e+00	1.244644e+03	-2.795864e+00	5.255712e-03

```
confint(modres)
```

```
Computing profile confidence intervals ...
```

	2.5 %	97.5 %
.sig01	380.58783843	413.09844224
.sig02	-0.21914874	-0.07662382
.sig03	53.63289793	61.16749962
.sigma	130.12879055	134.43896140
(Intercept)	-541.37831668	698.45608157
age	-3.37109955	3.34675337
sexM	-63.42013388	61.46926243
bmi	-6.47345513	2.10690550
time	-68.86489178	-21.06772501
exc1	-70.46863381	73.76138030
exc2	-72.75227356	76.99016534
exc3	-60.00518208	57.81173161
cbronchY	-48.24229385	49.26875300
smokingstatus	-28.41834884	73.89732242
pksm	-0.87395321	0.84909496
height	-3.20392403	3.41359668
emphysemaYes	-38.81061627	62.04642530
clushigh (> 0.2)	-143.50053348	30.65017150
bmi:time	0.03016077	1.57709036
time:smokingstatus	-25.50324319	-7.41358690
time:emphysemaYes	-17.20479674	0.78890078
time:clushigh (> 0.2)	-38.56464653	-6.80073844

```

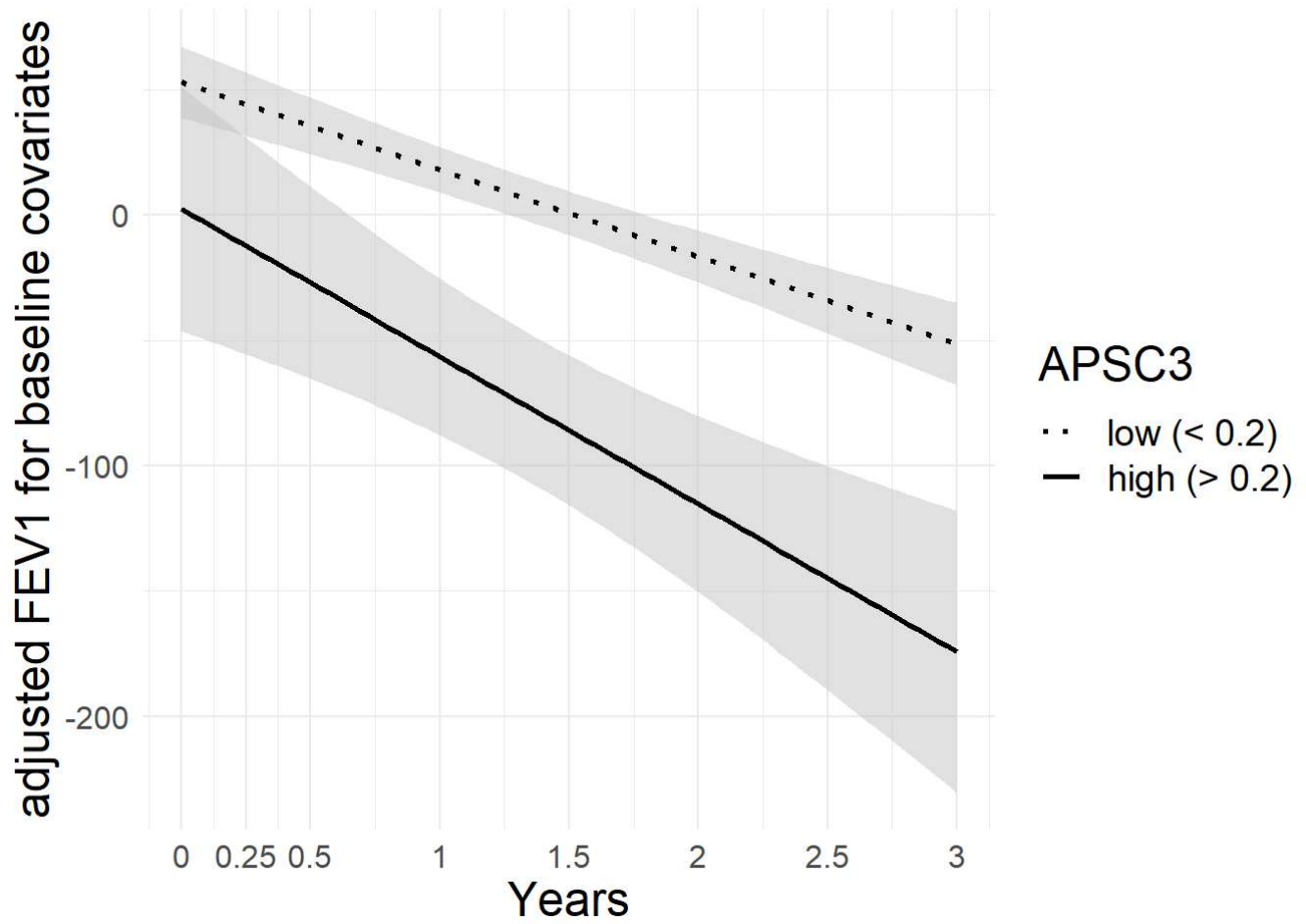
pred <- data.frame(dat, predicted=predict(modres))

plot1 <- ggplot(pred, aes(x = time, y = res, linetype = clus)) +
  geom_smooth(aes(fill = clus), method = "lm", se = TRUE, color = "black") +
  scale_linetype_manual(values = c("dotted", "solid"), name = "APSC3")+
  scale_fill_manual(values = c("grey", "grey"), guide = "none")+
  guides(
    linetype = guide_legend(
      override.aes = list(fill = "white"))
  ) +
  scale_x_continuous(breaks = c(0.0, 0.25, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0)
    , labels = c("0", "0.25", "0.5", "1", "1.5", "2", "2.5", "3"))+
  labs(x = "Years", y = "adjusted FEV1 for baseline covariates") +
  theme_minimal()+
  theme(
    axis.text = element_text(size = 12),
    axis.title = element_text(size = 18),
    plot.title = element_text(size = 18),
    legend.text = element_text(size = 14),
    legend.title = element_text(size = 18))

#png("intC6Adjusted.png", width = 2000, height = 1500, res = 300)
plot1

```

```
`geom_smooth()` using formula = 'y ~ x'
```



```
#dev.off()
```

By GOLD

```

t1 <- t
t2 <- t
t3 <- t

dat$clus <- NA
dat$clus[dat$goldcd==2 & dat$C6 < t1] <- "low"
dat$clus[dat$goldcd==2 & dat$C6 > t1] <- "high"

dat$clus[dat$goldcd==3 & dat$C6 < t2] <- "low"
dat$clus[dat$goldcd==3 & dat$C6 > t2] <- "high"

dat$clus[dat$goldcd==4 & dat$C6 < t3] <- "low"
dat$clus[dat$goldcd==4 & dat$C6 > t3] <- "high"

dat$clus <- factor(dat$clus, levels = c("low", "high"))

out <- c()
for(x in c(2,3,4)){
  mod <- lmer(
    fve ~ age + sex + bmi* time + exc + cbronch+
      smokingstatus* time+ pksm+ height+emphysema * time +
      clus*time+(1 + time| id),
    data = dat, subset = which(dat$goldcd==x))

  out <- rbind(out, summary(mod)$coeff["time:clushigh",])
}

```

Warning in checkConv(attr(opt, "derivs"), opt\$par, ctrl = control\$checkConv, :
Model failed to converge with max|grad| = 0.00245179 (tol = 0.002, component 1)

out

	Estimate	Std. Error	df	t value	Pr(> t)
[1,]	-24.147814	13.78321	587.6120	-1.7519732	0.08030003
[2,]	-27.089122	11.54642	505.9126	-2.3461063	0.01935576
[3,]	-8.350455	15.36110	138.4579	-0.5436105	0.58758376

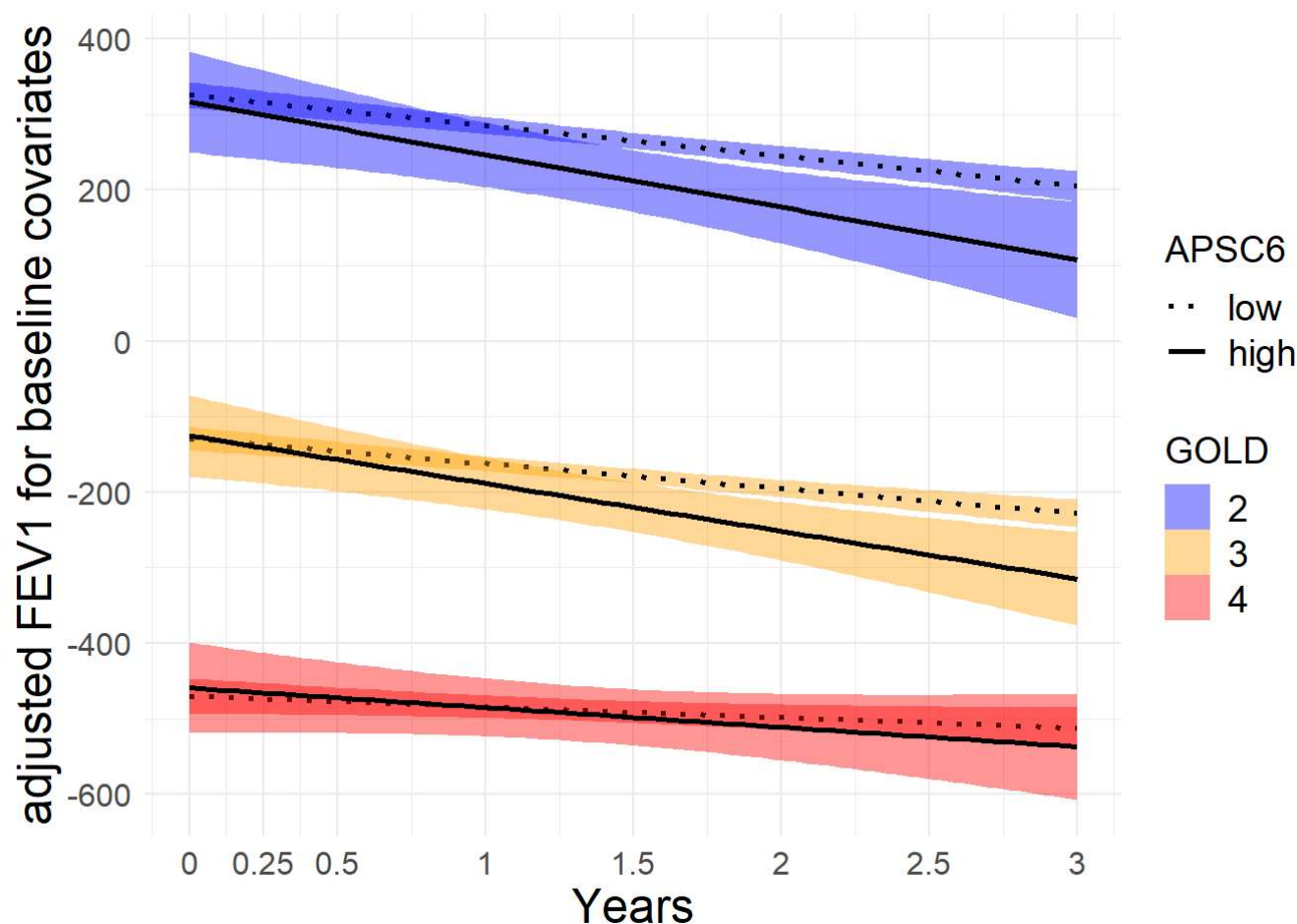
```

pred <- data.frame(dat, predicted=predict(modres))

plot1 <- ggplot(pred, aes(x = time, y = res, linetype = factor(clus))) +
  geom_smooth(aes(fill = factor(goldcd)), method = "lm", se = TRUE, color = "black") +
  scale_linetype_manual(values = c("dotted", "solid"), name = "APSC6")+
  scale_fill_manual(values = c("blue", "orange", "red"), name = "GOLD")+
  guides(
    linetype = guide_legend(
      override.aes = list(fill = "white")),
    fill = guide_legend(
      override.aes = list(linetype = "blank") # Add Line type to GOLD Legend
    )
  )+
  labs(x = "Years", y = "adjusted FEV1 for baseline covariates") +
  scale_x_continuous(breaks = c(0.0, 0.25, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0)
    , labels = c("0", "0.25", "0.5", "1", "1.5", "2", "2.5", "3")) +
  theme_minimal() +
  theme(legend.title = element_text(size = 14),
    axis.text = element_text(size = 12),
    axis.title = element_text(size = 18),
    plot.title = element_text(size = 18),
    legend.text = element_text(size = 14))
#png("intC6AdjustedByGOLD.png", width = 2000, height = 1500, res = 300)
plot1

```

`geom\_smooth()` using formula = 'y ~ x'



```
#dev.off()
```

Transcritomic analysis

Sputum

Load data:

```
load("./data/eset_sputum.rda")  
sputum_eset
```

```
ExpressionSet (storageMode: lockedEnvironment)  
assayData: 54675 features, 140 samples  
  element names: exprs  
protocolData: none  
phenoData  
  sampleNames: 30442 85931 ... 12897 (140 total)  
  varLabels: D_SUBJID AGE ... GROUP (83 total)  
  varMetadata: labelDescription  
featureData  
  featureNames: 1 2 ... 3 (54675 total)  
  fvarLabels: PROBEID ENSEMBL ... GENETYPE (9 total)  
  fvarMetadata: labelDescription  
experimentData: use 'experimentData(object)'  
Annotation: hgu133plus2
```

```
# Access to counts and phenotypical data  
norm_counts <- exprs(sputum_eset)  
  
features <- fData(sputum_eset)
```

```
# access to phenoData and change internal structure:  
df <- pData(sputum_eset)  
df <- df %>% mutate_at("D_SUBJID", as.character) %>%  
  mutate_at(c("SEX", "TRTGRP", "GOLDCD", "CBRONCH", "COUGH",  
              "SMOKER", "GROUP", "SUSMHS", "SUSMHSCD"),  
            as.factor) %>%  
  mutate_at(names(df[21:39]), as.factor)  
pData(sputum_eset) <- df
```

```

load("./data/model_data.rda")

clinicalData <- model_data

colData_prov <- pData(sputum_eset)

# merge both datasets by subject id
colData <- merge(colData_prov, clinicalData, by = "D_SUBJID",
                 all.x = TRUE)

colData <- colData %>% drop_na(c("AGE", "SEX", "BMI", "C2", "C3", "C6"))

# Access to count data for subjects that appear in clinicalData dataset:
countData <- norm_counts[,colData$D_SUBJID]

# Remove genes with NA
countData <- countData[complete.cases(countData),]

filter <- apply(countData, 1, function(x) mean(x>15)>0.8)
countData <- countData[filter,]

countData[countData<0] <-0

```

Sputum

APSC2

```

# perform two models
mod0 <- model.matrix(~AGE+centre+BMI+SEX,
                    data=colData)

mod_fev <- model.matrix(~C2+AGE+centre+BMI+SEX,
                    data=colData)

# apply the sva function to estimate the surrogate variables:
svseq = svaseq(countData,mod_fev,mod0)

```

Number of significant surrogate variables is: 18
 Iteration (out of 5):1 2 3 4 5

```

modSv = cbind(mod_fev,svseq$sv)

#fit the model
fit_sva_fev1 = lmFit(countData,modSv)
fit_sva_fev1 <- eBayes(fit_sva_fev1)
toptab <- topTable(fit_sva_fev1, coef="C2", adjust.method = "fdr")

```

```

genes <- features$SYMBOL
names(genes) <- features$PROBEID
toptab$GENES <- genes[rownames(toptab)]

write_xlsx(toptab, path="TableS3a.xlsx")

voom_fev1_res <- topTable(fit_sva_fev1,coef="C2",sort.by="p", number=nrow(fit_sva_fev1))

voom_fev1_res$GENES <- genes[rownames(voom_fev1_res)]

# select genes
mask_voom <- voom_fev1_res$P.Value < 0.05 & !is.na(voom_fev1_res$adj.P.Val)

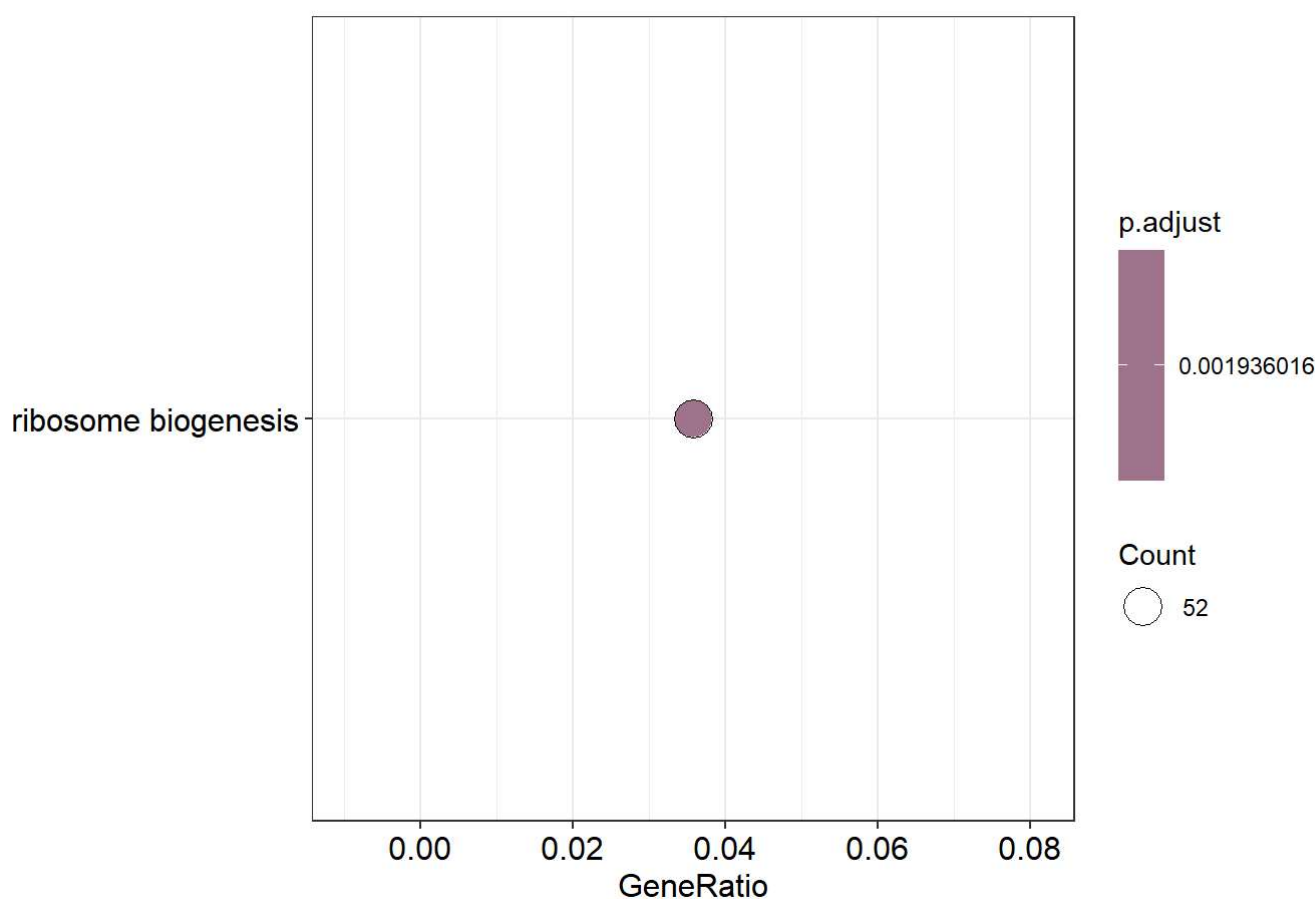
gns <- unique(unlist(strsplit(voom_fev1_res[mask_voom, "GENES"], "\\|")))

selgenes <- gns
ensembl <- useEnsembl(biomart = "genes", dataset = "hsapiens_gene_ensembl")
geneids<- getBM(c('hgnc_symbol','ensembl_gene_id','entrezgene_id','chromosome_name','start_position','end_position'), filters = 'hgnc_symbol', values=selgenes, mart=ensembl)
selgenes <- geneids[geneids$hgnc_symbol%in%selgenes, "entrezgene_id"]

#run enrichment in GO
GOMet1 <- enrichGO(gene = selgenes, 'org.Hs.eg.db', ont="BP", pvalueCutoff=0.05, pAdjustMethod="fdr")

clusterProfiler::dotplot(GOMet1)

```



```
as.data.frame(GOmet1)
```

ID	Description	GeneRatio	BgRatio	pvalue	p.a
<chr>	<chr>	<chr>	<chr>	<dbl>	
GO:0042254	GO:0042254 ribosome biogenesis	52/1452	325/18870	3.511092e-07	0.00193

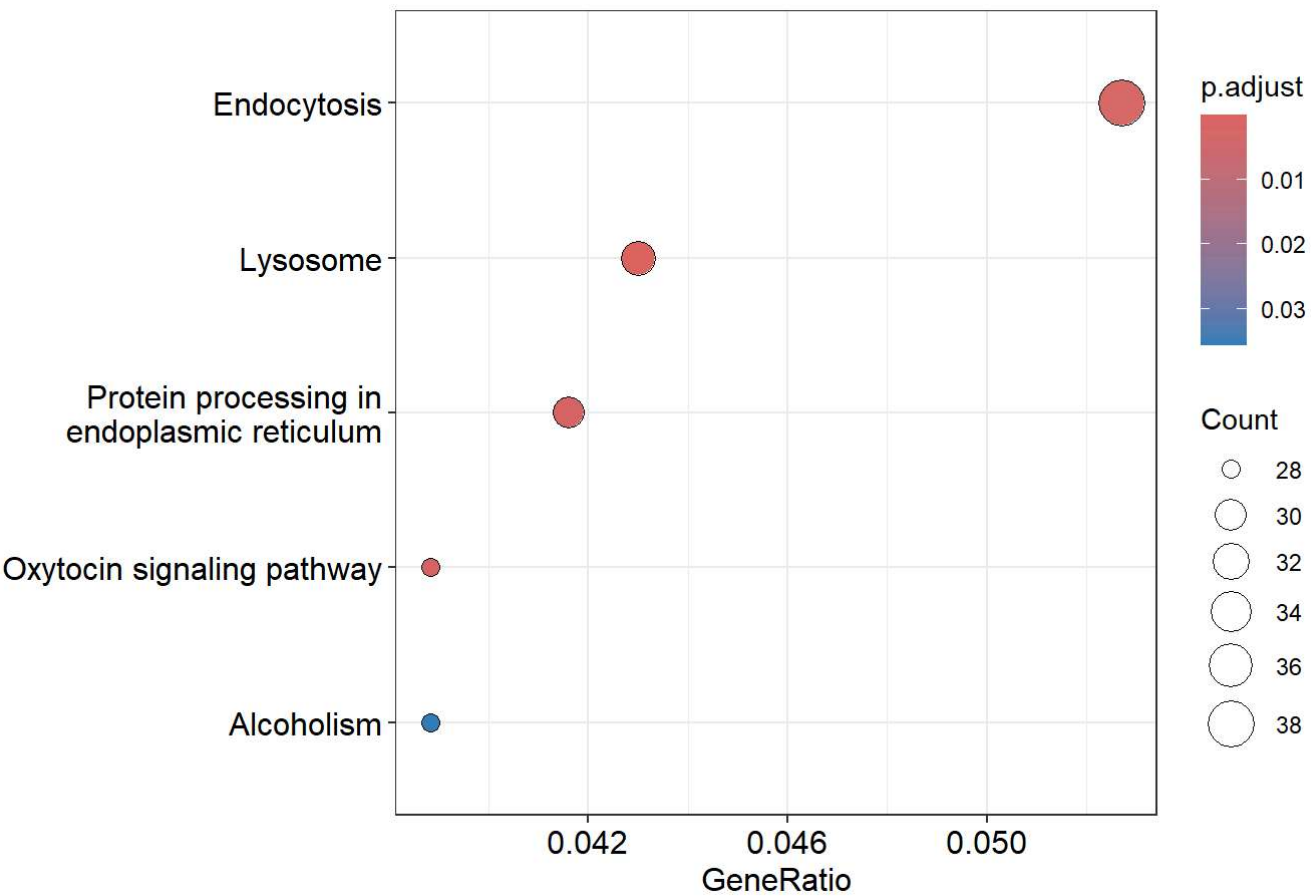
1 row | 1-8 of 10 columns

```
ans.kegg_v <- enrichKEGG(gene = selgenes,
                          organism = 'hsa',
                          pvalueCutoff = 0.05)
```

Reading KEGG annotation online: "https://rest.kegg.jp/link/hsa/pathway"...

Reading KEGG annotation online: "https://rest.kegg.jp/list/pathway/hsa"...

```
clusterProfiler::dotplot(ans.kegg_v)
```



```
as.data.frame(ans.kegg_v)
```

	category <chr>	subcategory <chr>	ID <chr>
hsa04142	Cellular Processes	Transport and catabolism	hsa041
hsa04921	Organismal Systems	Endocrine system	hsa049
hsa04141	Genetic Information Processing	Folding, sorting and degradation	hsa041
hsa04144	Cellular Processes	Transport and catabolism	hsa041
hsa05034	Human Diseases	Substance dependence	hsa050

5 rows | 1-4 of 12 columns

APSC3

```
# perform two models
mod0 <- model.matrix(~AGE+centre+BMI+SEX,
  data=countData)

mod_fev <- model.matrix(~C3+AGE+centre+BMI+SEX,
  data=countData)

# apply the sva function to estimate the surrogate variables:
svseq = svaseq(countData,mod_fev,mod0)
```

Number of significant surrogate variables is: 18
Iteration (out of 5):1 2 3 4 5

```
modSv = cbind(mod_fev,svseq$sv)

#fit the model
fit_sva_fev1 = lmFit(countData,modSv)
fit_sva_fev1 <- eBayes(fit_sva_fev1)
topTab <- topTable(fit_sva_fev1, coef="C3", adjust.method = "fdr")
```

APSC6

```
# perform two models
mod0 <- model.matrix(~AGE+centre+BMI+SEX,
  data=countData)

mod_fev <- model.matrix(~C6+AGE+centre+BMI+SEX,
  data=countData)

# apply the sva function to estimate the surrogate variables:
svseq = svaseq(countData,mod_fev,mod0)
```

Number of significant surrogate variables is: 18
Iteration (out of 5):1 2 3 4 5

```
modSv = cbind(mod_fev,svseq$sv)

#fit the model
fit_sva_fev1 = lmFit(countData,modSv)
fit_sva_fev1 <- eBayes(fit_sva_fev1)
toptab <- topTable(fit_sva_fev1, coef="C6", adjust.method = "fdr", sort.by="p", number=nrow(fit_sva_fev1))
```

```
genes <- features$SYMBOL
names(genes) <- features$PROBEID
toptab$GENES <- genes[rownames(toptab)]

# Create volcano plot
volcano_plot <- EnhancedVolcano(toptab,
                                lab = toptab$GENES,
                                x = 'logFC',
                                y = 'P.Value',
                                selectLab = toptab$GENES[1:25], # Top 10 genes by p-value
                                xlab = "Log2 Fold Change",
                                ylab = "-Log10 P-value",
                                pCutoff = 9.41E-5,
                                FCcutoff = 0,
                                pointSize = 2.5,
                                labSize = 4,
                                title = "APSC6",
                                subtitle = " ",
                                caption = "FDR-adjusted")

ggsave("volcano_plot.png", plot = volcano_plot, width = 10, height = 8, dpi = 300)
```

```

genes <- features$SYMBOL
names(genes) <- features$PROBEID
toptab$GENES <- genes[rownames(toptab)]

write_xlsx(toptab[voom_fev1_res$adj.P.Val < 0.05,], path="TableS3c.xlsx")

voom_fev1_res <- topTable(fit_sva_fev1,coef="C6",sort.by="p", number=nrow(fit_sva_fev1))

voom_fev1_res$GENES <- genes[rownames(voom_fev1_res)]

# select genes
mask_voom <- voom_fev1_res$adj.P.Val < 0.05 & !is.na(voom_fev1_res$adj.P.Val)

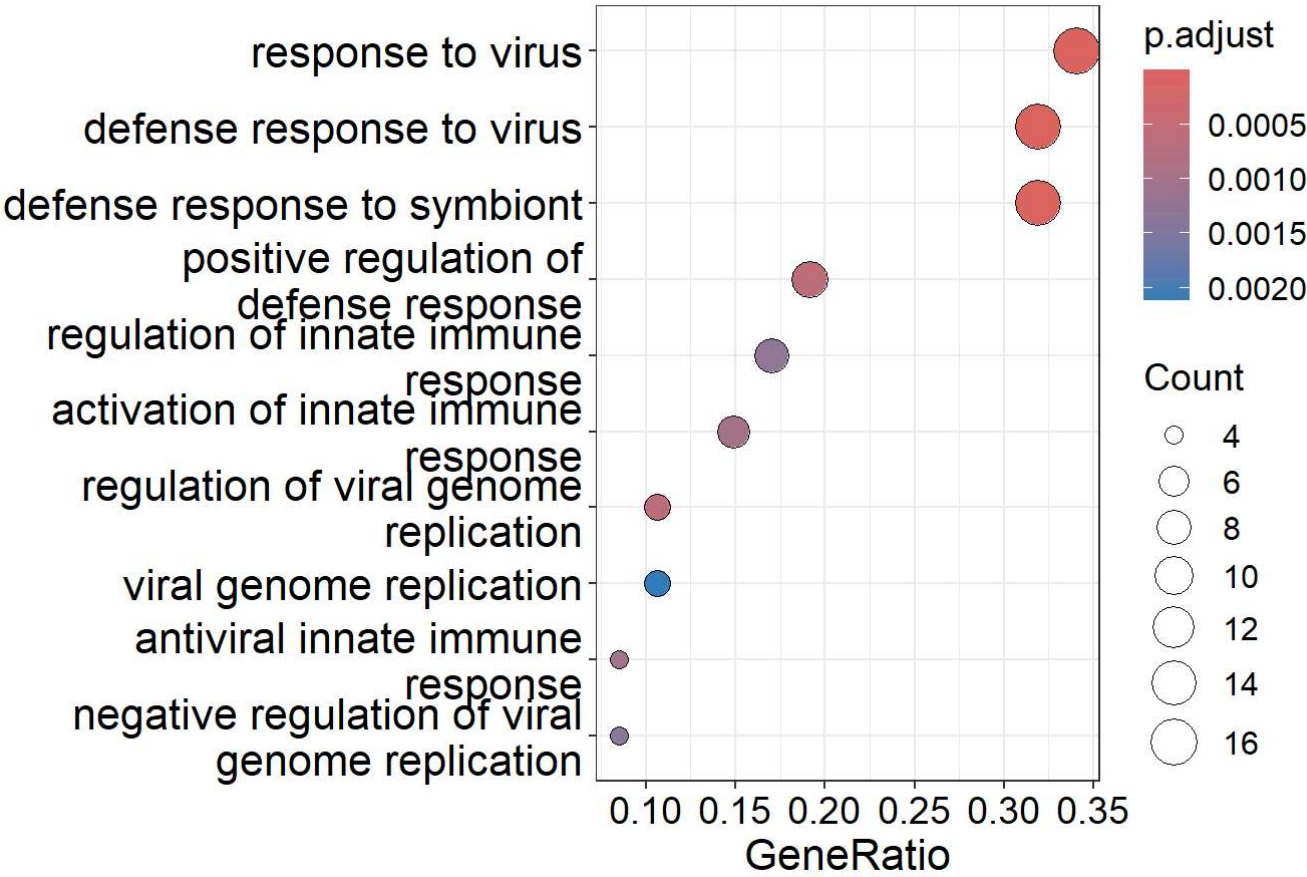
gns <- unique(unlist(strsplit(voom_fev1_res[mask_voom, "GENES"], "\\|")))

selgenes <- gns
ensembl <- useEnsembl(biomart = "genes", dataset = "hsapiens_gene_ensembl")
geneids<- getBM(c('hgnc_symbol','ensembl_gene_id','entrezgene_id','chromosome_name','start_position','end_position'), filters = 'hgnc_symbol', values=selgenes, mart=ensembl)
selgenes <- geneids[geneids$hgnc_symbol%in%selgenes, "entrezgene_id"]

#run enrichment in GO
GOMet1 <- enrichGO(gene = selgenes, 'org.Hs.eg.db', ont="BP", pvalueCutoff=0.05, pAdjustMethod="fdr")

#png("go.png", width = 2000, height = 1800, res = 300)
clusterProfiler::dotplot(GOMet1) +
  theme(axis.text.x = element_text(size = 14), # x-axis (e.g., GeneRatio)
        axis.text.y = element_text(size = 16), # y-axis (e.g., pathway names)
        axis.title = element_text(size = 16), # axis titles
        legend.text = element_text(size = 12), # legend labels
        legend.title = element_text(size = 14), # legend title
        plot.title = element_text(size = 18))

```



```
#dev.off()
```

```
as.data.frame(GOmet1)
```

	ID <chr>
GO:0051607	GO:0051607
GO:0140546	GO:0140546
GO:0009615	GO:0009615
GO:0045069	GO:0045069
GO:0031349	GO:0031349
GO:0002218	GO:0002218
GO:0140374	GO:0140374
GO:0045088	GO:0045088
GO:0045071	GO:0045071
GO:0019079	GO:0019079

1-10 of 56 rows | 1-2 of 10 columns

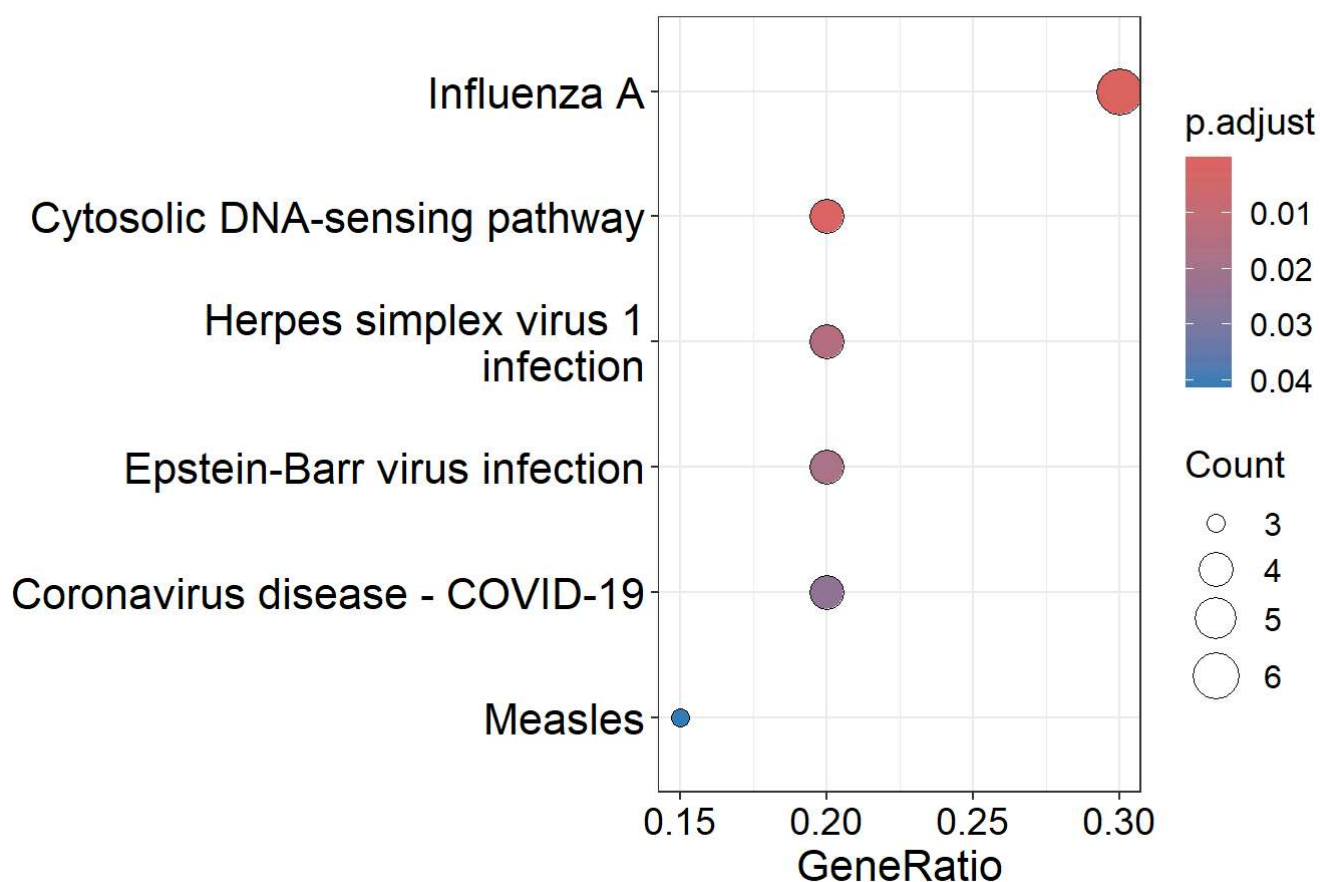
Previous 1 2 3 4 5 6 Next

```

ans.kegg_v <- enrichKEGG(gene = selgenes,
                        organism = 'hsa',
                        pvalueCutoff = 0.05)

#png("kegg.png", width = 2000, height = 1800, res = 300)
clusterProfiler::dotplot(ans.kegg_v) +
  theme(axis.text.x = element_text(size = 14), # x-axis (e.g., GeneRatio)
        axis.text.y = element_text(size = 16), # y-axis (e.g., pathway names)
        axis.title = element_text(size = 16), # axis titles
        legend.text = element_text(size = 12), # legend labels
        legend.title = element_text(size = 14), # legend title
        plot.title = element_text(size = 18))

```



```
dev.off()
```

```
null device
1
```

```
#as.data.frame(ans.kegg_v)
```

Associations between APSC6 and neutrophil counts

```
radcol <- colData(rdr_L1_final)
rownames(radcol) <- radcol$D_SUBJID
radcol <- radcol[colData$D_SUBJID,]

colData$NEUT_BLC.t1 <- radcol$NEUT_BLC.t1
colData$NEUT_BLC.t3 <- radcol$NEUT_BLC.t3
colData$NEUT_BLC.change <- radcol$NEUT_BLC.t3-radcol$NEUT_BLC.t1

colData$NEUT_BLQ.t1 <- radcol$NEUT_BLQ.t1
colData$NEUT_BLQ.t3 <- radcol$NEUT_BLQ.t3

summary(lm(colData$NEUT_BLC.change~colData$C6>0.2))
```

Call:

```
lm(formula = colData$NEUT_BLC.change ~ colData$C6 > 0.2)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.1448	-0.8223	0.0502	0.8702	4.0952

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.004796	0.157990	0.030	0.9758
colData\$C6 > 0.2TRUE	1.240204	0.575091	2.157	0.0333 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

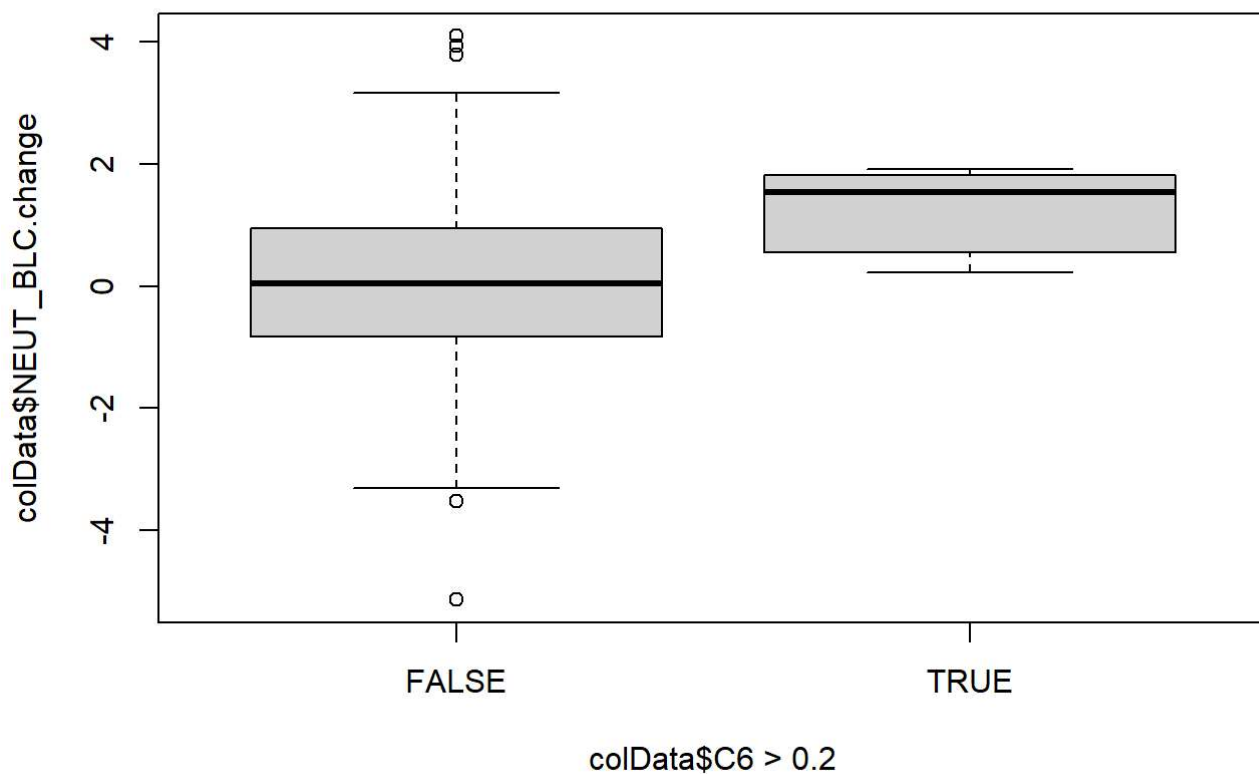
Residual standard error: 1.564 on 104 degrees of freedom

(19 observations deleted due to missingness)

Multiple R-squared: 0.0428, Adjusted R-squared: 0.0336

F-statistic: 4.651 on 1 and 104 DF, p-value: 0.03335

```
boxplot(colData$NEUT_BLC.change~colData$C6>0.2)
```



```

dat.neu <- dat.long
dat.neu <- dat.neu[dat.neu$time%in%c(0,3),]

neut <- c(colData$NEUT_BLQ.t1,colData$NEUT_BLQ.t3)
neutid <- paste(rep(c(0,3),each=nrow(colData)),c(colData$id,colData$id))

rownames(dat.neu) <- paste(dat.neu$time, dat.neu$id)
dat.neu <- dat.neu[neutid, ]

dat.neu$neut <- neut

dat.neu$clus <- factor(dat.neu$C6> 0.2, labels = c("low (< 0.2)", "high (> 0.2)"))

dat.neu <- dat.neu[!is.na(dat.neu$neut),]

sel <- names(table(dat.neu$id))[table(dat.neu$id)==2]

dat.neu <- dat.neu[dat.neu$id%in%sel,]

modres <- lmer(
  neut ~ age + sex + bmi + time*clus + (1| id),
  data = dat.neu)

summary(modres)

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]

Formula: neut ~ age + sex + bmi + time * clus + (1 | id)

Data: dat.neu

REML criterion at convergence: 1471.5

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.98244	-0.48058	-0.03243	0.50235	2.65628

Random effects:

Groups	Name	Variance	Std.Dev.
id	(Intercept)	39.79	6.308
Residual		35.59	5.966

Number of obs: 212, groups: id, 106

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	50.86907	10.35579	101.34266	4.912	3.47e-06 ***
age	0.17841	0.14439	101.00001	1.236	0.2195
sexM	1.53037	1.67500	101.00000	0.914	0.3631
bmi	0.05026	0.14576	101.00000	0.345	0.7310
time	0.48367	0.28410	104.00000	1.702	0.0917 .
clushigh (> 0.2)	-4.36852	3.37195	151.87434	-1.296	0.1971
time:clushigh (> 0.2)	1.97883	1.03414	104.00000	1.914	0.0584 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	age	sexM	bmi	time	c(>0.2)
age	-0.920					
sexM	0.272	-0.353				
bmi	-0.495	0.140	-0.157			
time	-0.041	0.000	0.000	0.000		
clshg(>0.2)	-0.290	0.183	-0.077	0.287	0.126	
tm:cl(>0.2)	0.011	0.000	0.000	0.000	-0.275	-0.460

confint(modres)

	2.5 %	97.5 %
.sig01	4.87733483	7.4668632
.sigma	5.19520055	6.8039202
(Intercept)	30.87584373	70.8622979
age	-0.10035722	0.4571778
sexM	-1.70346594	4.7642010
bmi	-0.23115333	0.3316673
time	-0.07290767	1.0402546
clushigh (> 0.2)	-10.87900187	2.1419691
time:clushigh (> 0.2)	-0.04715940	4.0048125

```

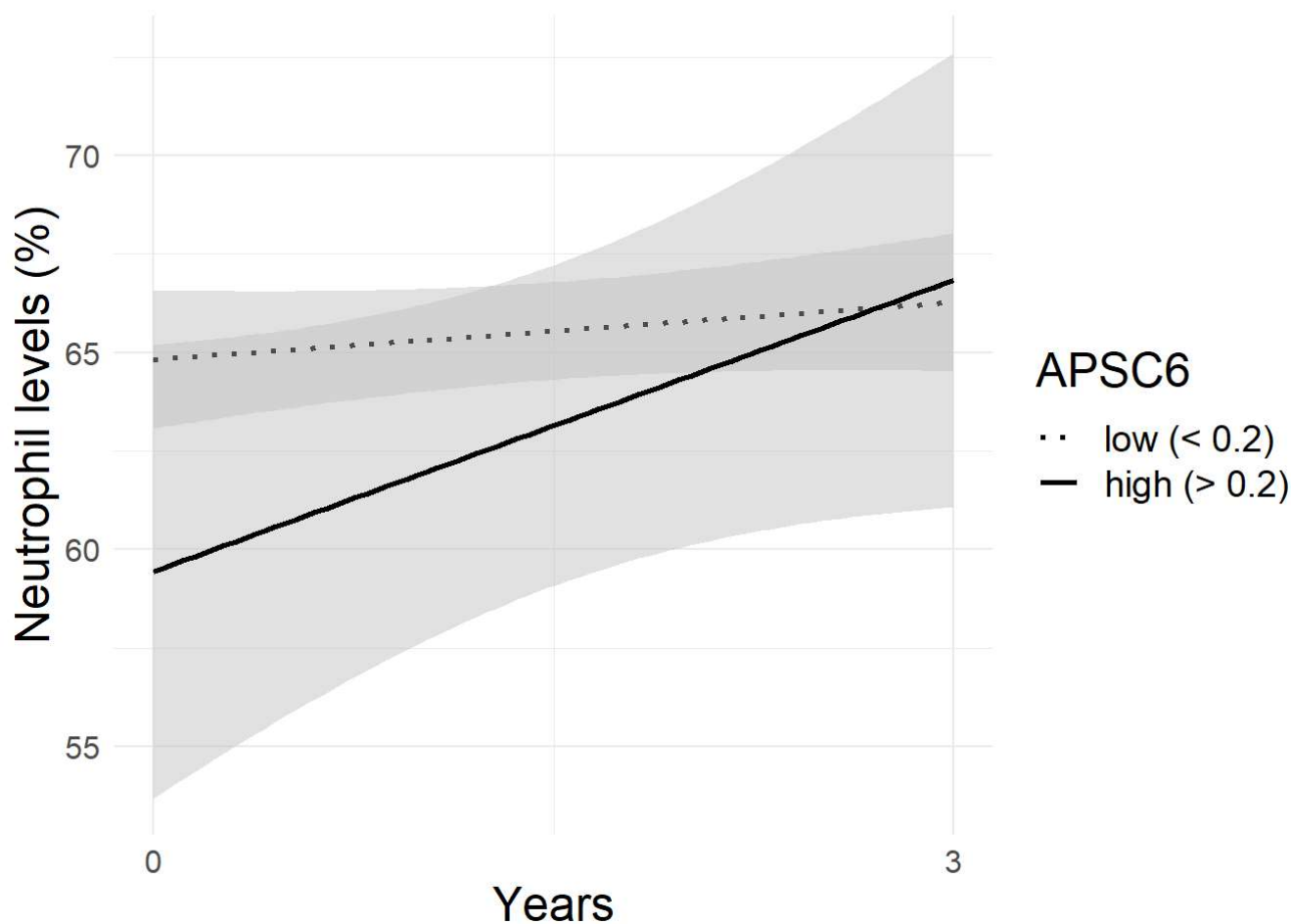
pred <- data.frame(dat.neu, predicted=predict(modres))

plot1 <- ggplot(pred, aes(x = time, y = neut, linetype = clus)) +
  geom_smooth(aes(fill = clus), method = "lm", se = TRUE, color = "black") +
  scale_linetype_manual(values = c("dotted", "solid"), name = "APSC6")+
  scale_fill_manual(values = c("grey", "grey"), guide = "none")+
  guides(
    linetype = guide_legend(
      override.aes = list(fill = "white")
    ) +
  scale_x_continuous(breaks = c(0.0, 3.0)
                    , labels = c("0", "3"))+
  labs(x = "Years", y = "Neutrophil levels (%)") +
  theme_minimal()+
  theme(
    axis.text = element_text(size = 12),
    axis.title = element_text(size = 18),
    plot.title = element_text(size = 18),
    legend.text = element_text(size = 14),
    legend.title = element_text(size = 18))

#png("neut.png", width = 2000, height = 1500, res = 300)
plot1

```

```
`geom_smooth()` using formula = 'y ~ x'
```



```
#dev.off()
```